

MULTIFRACTALITY IN ASSET RETURNS: THEORY AND EVIDENCE

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Abstract—This paper investigates the multifractal model of asset returns (MMAR), a class of continuous-time processes that incorporate the thick tails and volatility persistence exhibited by many financial time series. The simplest version of the MMAR compounds a Brownian motion with a multifractal time-deformation. Prices follow a semi-martingale, which precludes arbitrage in a standard two-asset economy. Volatility has long memory, and the highest finite moments of returns can take any value greater than 2. The local variability of a sample path is highly heterogeneous and is usefully characterized by the local Hölder exponent at every instant. In contrast with earlier processes, this exponent takes a continuum of values in any time interval. The MMAR predicts that the moments of returns vary as a power law of the time horizon. We confirm this property for Deutsche mark/U.S. dollar exchange rates and several equity series. We develop an estimation procedure and infer a parsimonious generating mechanism for the exchange rate. In Monte Carlo simulations, the estimated multifractal process replicates the scaling properties of the data and compares favorably with some alternative specifications.

I. Introduction

THE multifractal model of asset returns (MMAR) is a continuous-time process that captures the thick tails and long-memory volatility persistence that are exhibited by many financial time series.¹ It is constructed by compounding a standard Brownian motion with a random time-deformation process, which is specified to be multifractal. The time deformation produces clustering and long memory in volatility and implies that the moments of returns vary as a power law of the time horizon. We confirm this property empirically on the Deutsche mark/U.S. dollar exchange rate, a U.S. equity index, and several individual stocks.

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¹ Long memory is conveniently defined by hyperbolically declining autocorrelations either for a process itself or functions of it. This property was first analyzed in the context of fractional integration of Brownian motion by Mandelbrot (1965) and Mandelbrot and van Ness (1968). It has been documented in squared and absolute returns for many financial data sets (Taylor, 1986; Ding, Granger, & Engle, 1993; Dacorogna et al., 1993). Baillie (1996) provides a survey of long memory in economics.

The MMAR is characterized by a form of time-invariance called *multiscaling*, which combines extreme returns with long memory in volatility. This specification improves on traditional models with scaling features in several ways. First, the MMAR is consistent with economic equilibrium. The simplest version implies uncorrelated returns and semi-martingale prices, thus precluding arbitrage in a standard two-asset economy. The model also permits significant flexibility in matching the data. Returns have a finite variance, and their highest finite moment can take any value greater than 2. Consistent with many financial series, the unconditional distribution of returns displays thinner tails as the time scale increases. In contrast with earlier processes, however, the distribution need not converge to a Gaussian at low frequencies and never converges to a Gaussian at high frequencies. The MMAR thus captures the distributional nonlinearities that are exhibited by financial data, while retaining the parsimony and tractability of scaling models.

The time-deformation process is obtained as the limit of a simple iterative procedure called a *multiplicative cascade*. The construction begins with a uniform distribution of volatility at a suitably long time horizon, and randomly concentrates volatility into progressively smaller time intervals. The procedure follows the same rule at each stage of the cascade, which provides parsimony and implies moment scaling. By construction, volatility clusters at all frequencies, which is consistent with the intuition that economic factors such as technological shocks, business cycles, earnings cycles, and liquidity shocks have different time scales.² We anticipate that rational equilibrium models can generate the MMAR, either exogenously through multifractal shocks or endogenously due to market incompleteness or informational cascades.

The MMAR provides a fundamentally new class of stochastic processes to financial economists. In particular, the multifractal model is a continuous diffusion that lies outside the class of Itô processes. Although these traditional models locally vary as $(dt)^{1/2}$ along their sample paths, the MMAR generates the richer class $(dt)^{\alpha(t)}$, where the local scale $\alpha(t)$ can take a continuum of values. The relative occurrences of the local scales $\alpha(t)$ are conveniently summarized in a renormalized probability density called the *multifractal spectrum*. Given a specification of the model,

² This idea is further elaborated by Calvet and Fisher (2001).

we provide a general rule for calculating this function, and derive its closed-form expressions in a number of examples. The applied researcher can estimate the spectrum from the moments of the data and then infer the specification of the multifractal generating process.

Our empirical work begins by examining the Deutsche mark/U.S. dollar (DM/USD) exchange rate. We use a high-frequency data set of approximately 1.5 million quotes collected over one year, and a 23-year sample of daily prices. The exchange rate displays the moment-scaling property predicted by the model over a remarkable range of time horizons. We estimate the multifractal spectrum and infer a generating mechanism that replicates DM/USD scaling. Monte Carlo simulations then show that GARCH and FIGARCH are less likely to reproduce these results than the MMAR. In addition, we find evidence of scaling in a U.S. equity index and five individual stocks.³

Volatility modeling has received considerable attention in finance, and the most common approaches currently include numerous variants of the ARCH/GARCH class (Engle, 1982; Bollerslev, 1986) and stochastic volatility models (Wiggins, 1987).⁴ Because early processes in this literature had difficulty capturing the outliers of financial series, researchers have proposed conditional distributions of returns with thicker tails than the Gaussian. In discrete time, these adaptations include the Student- t (Bollerslev, 1987) and nonparametric specifications (Engle & Gonzalez-Rivera, 1991). The problem of modeling thick tails is more acute in continuous time and is typically addressed by incorporating an independent jump process. Bates (1995, 1996) thus finds that standard diffusions cannot produce tails sufficiently fat to explain the implied volatility smile in option prices, and recommends the incorporation of jumps. Although a continuous diffusion, the MMAR incorporates enough bursts of extreme volatility to capture the fat tails of financial series. It also extends the characterization of volatility in continuous time by considering a multiplicity of local scales. In particular, the multifractal model can generate local oscillations that are intermediate between Itô diffusion and discontinuous jumps.

Although early processes from the ARCH/GARCH literature have weak persistence, long memory in squared returns is a characteristic feature of FIGARCH (Baillie, Bollerslev, & Mikkelsen, 1996) and the Long Memory Stochastic Volatility (LMSV) approach (Breidt, Crato, &

DeLima, 1998). The MMAR is reminiscent of the long-memory property of these models. In addition, the multifractal process has appealing temporal aggregation properties and is parsimoniously consistent with the moment scaling of financial data.

Subsection IA discusses the relation between the MMAR and earlier scaling models. Section II defines multifractals and demonstrates their construction through a number of simple examples. Section III formalizes the MMAR by compounding a Brownian motion with a continuous time-deformation process. Section IV shows that multifractal processes can take a continuum of local scales, whose distribution is conveniently characterized by the multifractal spectrum. Section V extends the model to permit long memory in returns, which allows testing of the martingale hypothesis and may be useful in modeling economic series with persistence. In section VI, we verify the moment-scaling rule for DM/USD exchange rates and estimate the corresponding multifractal spectrum. We infer a data-generating process and show that simulated samples replicate the scaling features of the data. Evidence of multifractal scaling is also found in a U.S. equity index and five individual stocks. Section VII summarizes our results and discusses possible extensions.

This paper simplifies the discussion and extends the results of three earlier working papers (Mandelbrot, Fisher, & Calvet, 1997; Calvet, Fisher, & Mandelbrot, 1997; Fisher, Calvet, & Mandelbrot, 1997). In the remainder of the text, we refer to the working papers as *MFC*, *CFM*, and *FCM*, respectively, signifying the various permutations of the authors. All proofs are contained in the appendix.

A. Review of the Literature

The multifractal model combines several elements of previous research on financial time series. First, the MMAR generates fat tails in the unconditional distribution of returns, and is thus reminiscent of the L -stable processes of Mandelbrot (1963).⁵ The MMAR improves on this earlier model by generating returns with a finite variance, as seems to be empirically the case in most financial series. Furthermore, the L -stable model assumes that increments are independent through time, and have thus the same variability at every instant. In contrast, the MMAR helps model one of the main features of financial markets: fluctuations in volatility.

Second, the multifractal model has long memory in the absolute value of returns, but the returns themselves have a white spectrum. Long memory is the characteristic feature of fractional Brownian motion (FBM), introduced by Kolmogorov (1940) and Mandelbrot (1965). An FBM, denoted $B_H(t)$, has continuous sample paths, as well as Gaussian and possibly dependent increments. The FBM is an ordinary

³ The moment-scaling properties of financial returns are also the object of a growing physics literature (Galluccio et al., 1997; Vandewalle & Ausloos, 1998; Pasquini & Serva, 2000; Richards, 2000). These contributions confirm that multiscaling is exhibited by many financial time series and are thus complementary of the empirical work contained in Fisher, Calvet, and Mandelbrot (1997) and further developed in this paper. Although the physics literature focuses on these phenomenological regularities, the MMAR is a parsimonious stochastic process that allows a unified treatment of the theoretical and empirical properties of the price dynamics.

⁴ See Ghysels, Harvey, and Renault (1996) for a recent survey of the stochastic volatility literature.

⁵ Recent applications of the L -stable model to foreign exchange rates include Koedijk and Kool (1992) and Phillips, McFarland, and McMahon (1996).

Brownian motion for $H = 1/2$, is antipersistent when $0 < H < 1/2$, and displays persistence and long memory when $1/2 < H < 1$. Granger and Joyeux (1980) and Hosking (1981) introduced ARFIMA, a discrete-time counterpart of the FBM that helped the use of long memory in economics. FBM and ARFIMA do not disentangle volatility persistence from long memory in returns.⁶ This has obvious limitations in financial applications and has led to the construction of processes such as FIGARCH and LMSV. Like these recent models, the MMAR separates persistence in volatility from persistence in returns, but in a parsimonious, continuous-time setting.

The third essential component of the multifractal model is the concept of trading time.

Definition 1: Let $\{B(t)\}$ be a stochastic process, and $\theta(t)$ an increasing function of t . We call $X(t) \equiv B[\theta(t)]$ a compound⁷ process. The index t denotes clock time, and $\theta(t)$ is called the *trading time* or *time-deformation process*.

When the directing process B is a martingale, the trading time speeds up or slows down the process $X(t)$ without influencing its direction. Compounding can thus separate the direction and the size of price movements, and has been used in the literature to model the unobserved natural time scale of economic series (Mandelbrot & Taylor, 1967; Clark, 1973; Stock, 1987, 1988). More recently, this method has been used to build models integrating seasonal factors (Dacorogna et al., 1993; Müller et al., 1995) and measures of market activity (Ghysels, Gouriéroux, & Jasiak, 1996). The MMAR also incorporates compounding, and its primary innovation is to specify the trading time θ to be multifractal. Although the MMAR is not a structural model of trade, future work may define the trading time θ to be a function of observable data.

Finally, the MMAR generalizes the concept of scaling, in the sense that a well-defined rule relates returns over different sampling intervals. Mandelbrot (1963) suggested that the shape of the distribution of returns should be the same when the time scale is changed, or, more formally:

Definition 2: A random process $\{X(t)\}$ that satisfies

$$\{X(ct_1), \dots, X(ct_k)\} \stackrel{d}{=} \{c^H X(t_1), \dots, c^H X(t_k)\}$$

for some $H > 0$ and all $c, k, t_1, \dots, t_k \geq 0$, is called *self-affine*.⁸ The number H is the *self-affinity index*, or *scaling exponent*, of the process $\{X(t)\}$.

⁶ Taqqu (1975) establishes that $B_H(t)$ has long memory in the absolute value of increments when $H > 1/2$.

⁷ Processes of this type have also been called *subordinated* in the recent economics literature. In mathematics, subordination differs from compounding, and requires that $\theta(t)$ have independent increments (Bochner, 1955; Feller, 1968). The economics literature has evolved to describe any generic time-deformation process as a subordinator.

⁸ Self-affine processes are sometimes called *self-similar* in the literature.

The Brownian motion, the L -stable process, and the FBM are the main examples of self-affine processes in finance. Empirical evidence suggests that many financial series are not self-affine, but instead have thinner tails and become less peaked in the bells when the sampling interval increases. The MMAR captures this feature, as well as a generalized version of self-affinity exhibited by the data. While maintaining the simplicity of self-affine processes, the MMAR is thus sufficiently flexible to model the nonlinearities, fat tails, and long-memory volatility persistence exhibited by many financial time series.

II. Multifractal Measures and Processes

The MMAR is constructed in section III by compounding a Brownian motion, $B(t)$, with a random increasing function, $\theta(t)$:

$$\ln P(t) - \ln P(0) = B[\theta(t)].$$

The trading time $\theta(t)$ will be specified as the cumulative distribution function (c.d.f.) of a multifractal measure μ , a concept that we now present.

A. The Binomial Measure

Multifractal measures can be built by iterating a simple procedure called a *multiplicative cascade*. We first present one of the simplest examples, the binomial measure⁹ on $[0, 1]$. Consider the uniform probability measure μ_0 on the unit interval, and two positive numbers m_0 and m_1 adding up to 1. In the first step of the cascade, we define a measure μ_1 by uniformly spreading the mass m_0 on the left subinterval $[0, 1/2]$, and the mass m_1 on the right subinterval $[1/2, 1]$. The density of μ_1 is a step function, as illustrated in figure 1a.

In the second stage of the cascade, we split the interval $[0, 1/2]$ into two subintervals of equal length. The left subinterval $[0, 1/4]$ is allocated a fraction m_0 of $\mu_1[0, 1/2]$, whereas the right subinterval $[1/4, 1/2]$ receives a fraction m_1 . Applying a similar procedure to $[1/2, 1]$, we obtain a measure μ_2 such that

$$\mu_2[0, 1/4] = m_0 m_0, \quad \mu_2[1/4, 1/2] = m_0 m_1,$$

$$\mu_2[1/2, 3/4] = m_1 m_0, \quad \mu_2[3/4, 1] = m_1 m_1.$$

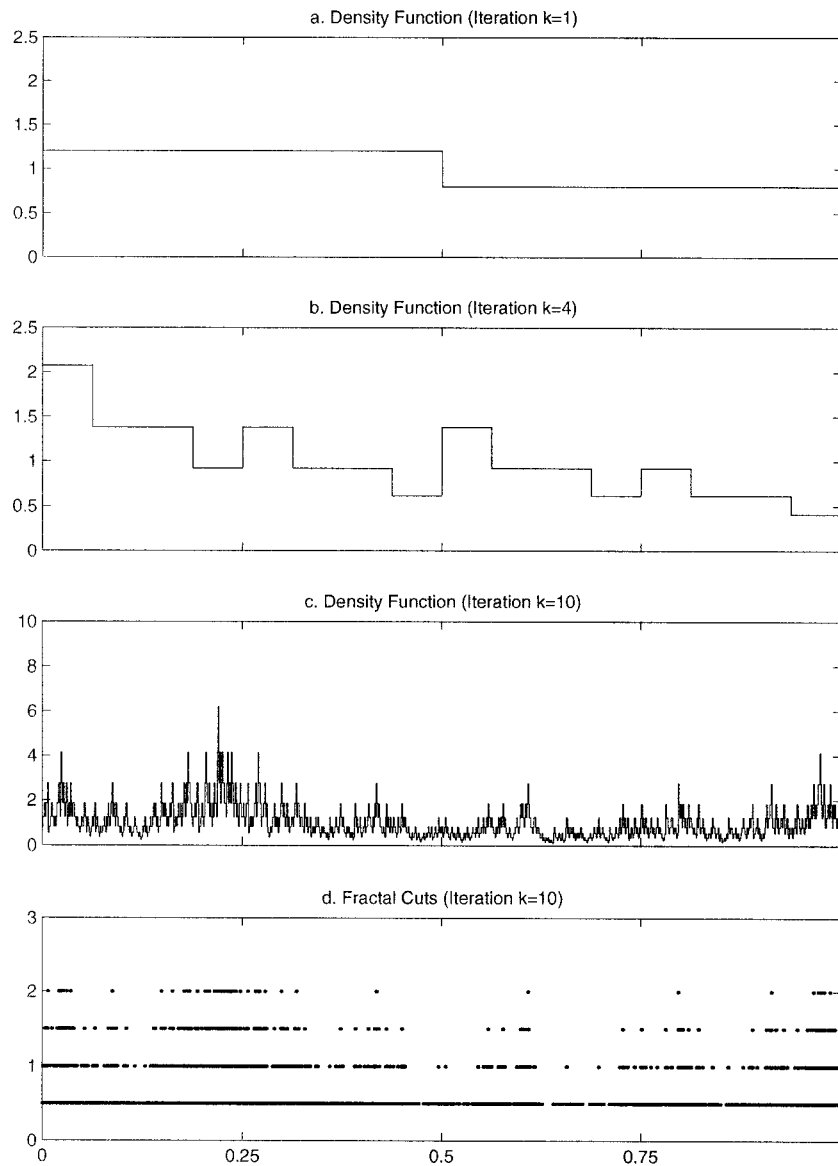
Iteration of this procedure generates an infinite sequence of measures (μ_k) that weakly converges to the binomial measure μ . Figure 1b illustrates the density of the measure μ_4 obtained after $k = 4$ steps of the recursion.

Because $m_0 + m_1 = 1$, each stage of the construction preserves the mass of split dyadic intervals.¹⁰ Consider the

⁹ The binomial is sometimes called the *Bernoulli* or *Besicovitch measure*.

¹⁰ A number $t \in [0, 1]$ is called *dyadic* if $t = 1$ or $t = \eta_1 2^{-1} + \dots + \eta_k 2^{-k}$ for a finite k and $\eta_1, \dots, \eta_k \in \{0, 1\}$. A dyadic interval has dyadic endpoints.

FIGURE 1.—CONSTRUCTION OF THE BINOMIAL MEASURE



In panels (a) and (b), the construction is deterministic with the fraction $m_0 = 0.6$ of the mass always allocated to the left, and fraction $m_1 = 0.4$ always allocated to the right. Panel (c) shows a randomized binomial measure after $k = 10$ stages. The masses m_0 and m_1 each have equal probabilities of going to the left or right. The final panel shows the fractal character of "cuts" of various sizes. Each cut shows the set of instants at which the random measure in panel (c) exceeds a given level. The clustering of these sets has a self-similar structure, and the extreme bursts of volatility are intermittent, as discussed in the appendix (subsection F).

interval $[t, t + 2^{-k}]$, where $t = \overline{0 \cdot \eta_1 \cdots \eta_k} = \sum_{i=1}^k \eta_i 2^{-i}$ for some $\eta_1, \dots, \eta_k \in \{0, 1\}$. We denote by φ_0 and φ_1 the relative frequencies of 0's and 1's in $(\eta_1, \dots, \eta_k)$. The measure of the dyadic interval then simplifies to $\mu[t, t + 2^{-k}] = m_0^{k\varphi_0} m_1^{k\varphi_1}$. This illustrates that the construction creates large and increasing heterogeneity in the allocation of mass. As a result, the binomial, like many multifractals, is a continuous but singular probability measure that has no density and no point mass.

This construction is easily generalized. For instance, we can uniformly split intervals into an arbitrary number $b \geq 2$ of cells at each stage of the cascade. Subintervals, indexed from left to right by β ($0 \leq \beta \leq b - 1$), then receive fractions $m_0, \dots, m_{b-1}$ of the measure. We preserve mass

in the construction by imposing that these fractions, also called *multipliers*, add up to 1: $\sum m_\beta = 1$. This defines the class of multinomial measures, which are discussed by Mandelbrot (1989a).

A more significant extension randomizes the allocation of mass between subintervals. The multiplier of each cell is then a discrete random variable, M_β , taking values $m_0, m_1, \dots, m_{b-1}$ with probabilities $p_0, \dots, p_{b-1}$. We preserve mass in the construction by imposing the additivity constraint: $\sum M_\beta = 1$. This modified algorithm generates a random multifractal measure. Figure 1c shows the random density obtained after $k = 10$ iterations with parameters $b = 2$, $p = p_0 = 0.5$, and $m_0 = 0.6$. This density, which represents the flow of trading time, begins to show the

properties that we desire in modeling financial volatility. The occasional bursts of trading time generate thick tails in the compound price process, and their clustering generates volatility persistence. Because the reshuffling of mass follows the same rule at each stage of the cascade, volatility clustering is present at all time scales.

B. Multiplicative Measures

We can also consider nonnegative multipliers, M_β ($0 \leq \beta \leq b - 1$), with arbitrary distributions. Assume for simplicity that all multipliers are identically distributed ($M_\beta \stackrel{d}{=} M \forall \beta$) and that multipliers defined at different stages of the construction are independent. The limit multiplicative measure is called *conservative* when mass is conserved exactly at each stage ($\sum M_\beta \equiv 1$), and *canonical* when it is preserved only *on average* ($\mathbb{E}(\sum M_\beta) = 1$, or, equivalently, $\mathbb{E}M = 1/b$). A canonical measure can be conveniently generated by choosing independent multipliers, M_β , within each stage of the cascade.

The moments of multiplicative measures have interesting scaling properties. To show this, first consider the generating cascade of a conservative measure, μ . Stage 1 uniformly splits the unit interval into cells of length b^{-1} and allocates random masses, $M_0, \dots, M_{b-1}$, to each cell. Similarly, the measure of a b -adic cell of length $\Delta t = b^{-k}$, starting at $t = 0 \cdot \eta_1 \dots \eta_k = \sum \eta_i b^{-i}$, is the product of k multipliers:

$$\mu(\Delta t) = M_{\eta_1} M_{\eta_1, \eta_2} \dots M_{\eta_1, \dots, \eta_k} \quad (1)$$

Because multipliers defined at different stages of the cascade are independent, we infer that $\mathbb{E}[\mu(\Delta t)^q] = [\mathbb{E}(M^q)]^k$, or, equivalently:

$$\mathbb{E}[\mu(\Delta t)^q] = (\Delta t)^{\tau(q)+1}, \quad (2)$$

where $\tau(q) = -\log_b \mathbb{E}(M^q) - 1$. The moment of an interval's measure is thus a power functions of the length Δt . This important scaling rule characterizes multifractals.

Scaling relation (2) easily generalizes to a canonical measure μ , which by definition is generated by a cascade that only conserves mass on average: $\mathbb{E}(\sum M_\beta) = 1$. The mass of the unit interval is then a random variable $\Omega = \mu[0, 1] \geq 0$. More generally, the measure of a b -adic cell satisfies

$$\mu(\Delta t) = M_{\eta_1} M_{\eta_1, \eta_2} \dots M_{\eta_1, \dots, \eta_k} \Omega_{\eta_1, \dots, \eta_k}, \quad (3)$$

where $\Omega_{\eta_1, \dots, \eta_k}$ has the same distribution as Ω . This directly implies the scaling relationship

$$\mathbb{E}[\mu(\Delta t)^q] = \mathbb{E}(\Omega^q)(\Delta t)^{\tau(q)+1}, \quad (4)$$

which generalizes equation (2).

The right tail of the measure $\mu(\Delta t)$ is determined by the way mass is preserved at each stage of the construction.

When μ is conservative, the mass of the cell is bounded above by the deterministic mass of the unit interval: $0 \leq \mu(\Delta t) \leq \mu[0, 1] = 1$, and has therefore finite moments of every order. On the other hand, consider a canonical measure generated by independent multipliers, M_β . We assume for simplicity that $\mathbb{E}(M^q) < \infty$ for all q . Guivarc'h (1987) shows that the random mass $\Omega \geq 0$ of the unit interval then has a Paretian right tail:

$$\mathbb{P}\{\Omega > \omega\} \sim C_1 \omega^{-q_{crit}} \text{ as } \omega \rightarrow +\infty,$$

where $C_1 > 0$ and the critical moment, q_{crit} , is finite and larger than 1: $1 < q_{crit} < \infty$.¹¹ By equation (3), the mass of every cell has the same critical moment, q_{crit} , as the random variable Ω . The property $q_{crit} > 1$ will prove particularly important because it implies that returns have a finite variance in the MMAR.

The multiplicative measures constructed so far are grid-bound, in the sense that the scaling rule of equation (4) holds only when $t = 0 \cdot \eta_1, \dots, \eta_k$ and $\Delta t = b^{-l}$, $l \geq k$. Let $\mathcal{D}$ denote the set of couples $(t, \Delta t)$ satisfying scaling rule (4). $\mathcal{D}$ has interesting topological properties that are summarized in Property 1 of the appendix (subsection A). Alternatively, we can consider grid-free random measures that satisfy scaling rule (4) for all admissible values of $(t, \Delta t)$ (Mandelbrot, 1989a). This leads to the following.

Definition 3: A random measure μ defined on $[0, 1]$ is called *multifractal* if it satisfies

$$\mathbb{E}(\mu[t, t + \Delta t]^q) = c(q)(\Delta t)^{\tau(q)+1}$$

$$\text{for all } (t, \Delta t) \in \mathcal{D}, q \in \mathcal{Q},$$

where $\mathcal{D}$ is a subset of $[0, 1] \times [0, 1]$, $\mathcal{Q}$ is an interval, and $\tau(q)$ and $c(q)$ are functions with domain $\mathcal{Q}$. Moreover, $[0, 1] \subseteq \mathcal{Q}$, and $\mathcal{D}$ satisfies Property 1, which is defined in the appendix.

Maintaining the distinction between grid-bound and grid-free measures would prove cumbersome and lead to unnecessary technicalities. We will therefore neglect the difference between the two classes in the remainder of this paper. The interested reader can refer to Calvet and Fisher (2001) for a detailed treatment of grid-free multifractals.

C. Multifractal Processes

Multifractality is easily extended from measures to stochastic processes:

Definition 4: A stochastic process $\{X(t)\}$ is called *multifractal* if it has stationary increments and satisfies

¹¹ The cascade construction also implies that Ω satisfies the invariance relation $\sum_{i=1}^b M_i \Omega_i \stackrel{d}{=} \Omega$, where $M_1, \dots, M_b, \Omega_1, \dots, \Omega_b$ are independent copies of the random variables M and Ω .

$$\mathbb{E}(|X(t)|^q) = c(q)t^{\tau(q)+1}, \quad \text{for all } t \in \mathcal{T}, q \in \mathcal{Q}, \quad (5)$$

where $\mathcal{T}$ and $\mathcal{Q}$ are intervals on the real line, $\tau(q)$ and $c(q)$ are functions with domain $\mathcal{Q}$. Moreover, $\mathcal{T}$ and $\mathcal{Q}$ have positive lengths, and $0 \in \mathcal{T}$, $[0, 1] \subseteq \mathcal{Q}$.

The function $\tau(q)$ is called the *scaling function* of the multifractal process. Setting $q = 0$ in condition (5), we see that all scaling functions have the same intercept $\tau(0) = -1$. In addition, it is easy to show

Proposition 1: The scaling function $\tau(q)$ is concave.

We will see that the distinction between linear and non-linear scaling functions $\tau(q)$ is particularly important.

A self-affine process $\{X(t)\}$ is multifractal and has a linear function $\tau(q)$, as is now shown. Denoting by H the self-affinity index, we observe that the invariance condition $X(t) \stackrel{d}{=} t^H X(1)$ implies that $\mathbb{E}(|X(t)|^q) = t^{Hq} \mathbb{E}(|X(1)|^q)$. Scaling rule (5) therefore holds with $c(q) = \mathbb{E}(|X(1)|^q)$ and $\tau(q) = Hq - 1$. In this special case, the scaling function $\tau(q)$ is linear and fully determined by its index H . More generally, linear scaling functions $\tau(q)$ are determined by a unique parameter (their slope) and the corresponding processes are called *uniscaling* or *unifractal*.

Uniscaling processes, which may seem appealing for their simplicity, do not satisfactorily model asset returns. This is because most financial data sets have thinner tails and become less peaked in the bell when the sampling intervals, Δt , increases. In this paper, we focus on multiscaling processes, which have a nonlinear $\tau(q)$. They provide a parsimonious framework with strict moment conditions and enough flexibility to model a wide range of financial prices.

III. The Multifractal Model of Asset Returns

We now formalize construction of the MMAR. Consider the price of a financial asset $P(t)$ on a bounded interval $[0, T]$, and define the log-price process

$$X(t) \equiv \ln P(t) - \ln P(0).$$

We model $X(t)$ by compounding a Brownian motion with a multifractal trading time:

Assumption 1: $X(t)$ is a compound process

$$X(t) \equiv B[\theta(t)],$$

where $B(t)$ is a Brownian motion, and $\theta(t)$ is a stochastic trading time.

Assumption 2: Trading time $\theta(t)$ is the c.d.f. of a multifractal measure μ defined on $[0, T]$.

Assumption 3: The processes $\{B(t)\}$ and $\{\theta(t)\}$ are independent.

This construction generates a large class of multifractal processes.

We will show that the price process is a semi-martingale, which implies the absence of arbitrage in simple economies. A straightforward generalization of this model allows the broader class of fractional Brownians $B_H(t)$ in Assumption 1, as developed in section V. In Assumption 2, the multifractal measure μ can be multinomial or multiplicative, which implies a continuous trading time $\theta(t)$ with nondecreasing paths and stationary increments. Assumption 3 ensures that the unconditional distribution of returns is symmetric. Weakening this assumption allows leverage effects, as in Nelson (1991) and Glosten, Jagannathan, and Runkle (1993), and is a promising direction for future research.

Under the previous assumptions,

Theorem 1: The log-price $X(t)$ is a multifractal process with stationary increments and scaling function $\tau_X(q) \equiv \tau_\theta(q/2)$.

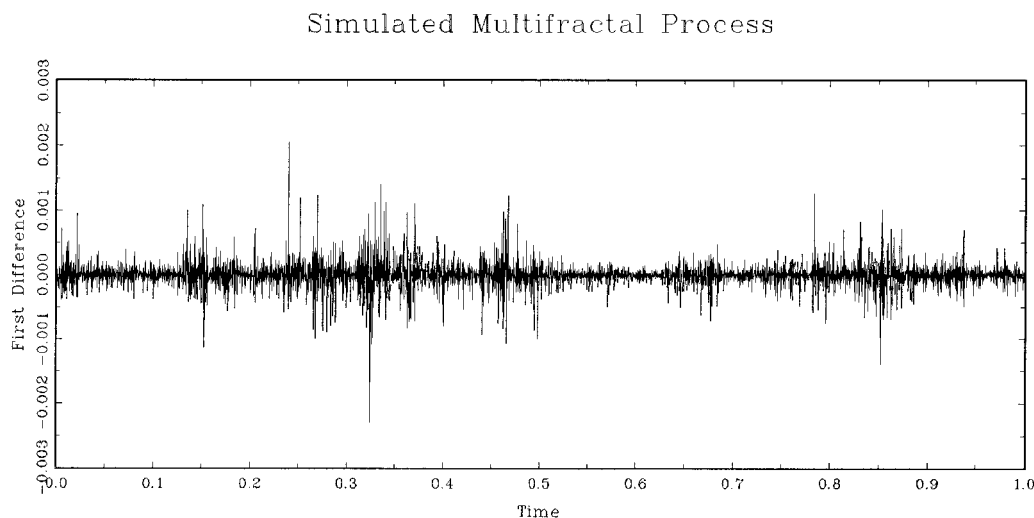
Trading time controls the tails of the process $X(t)$. As shown in the proof, the q^{th} moment of X exists if (and only if) the process θ has a moment of order $q/2$. In particular, if $\mathbb{E}|X(t)|^q$ is finite for some instant t , then it is finite for all t . We therefore drop the time index when discussing the critical moment of the multifractal process.

The tails of $X(t)$ have different properties if the generating measure is conservative or canonical. This follows directly from the discussion of subsection IIA. If μ is conservative, trading time is bounded and the process $X(t)$ has finite moments of all (nonnegative) order. Conservative measures thus generate “mild” processes with relatively thin tails. Conversely, the total mass $\theta(T) \equiv \mu[0, T]$ of a canonical measure is a random variable with Paretian tails. In particular, there exists a critical exponent $q_{\text{crit}}(\theta) > 1$ for trading time such that $\mathbb{E}\theta^q$ is finite when $0 \leq q < q_{\text{crit}}(\theta)$ and infinite when $q \geq q_{\text{crit}}(\theta)$.¹² The log-price $X(t)$ then has infinite moments and is accordingly called “wild.” Note, however, that $X(t)$ always has finite variance, since $q_{\text{crit}}(X) = 2q_{\text{crit}}(\theta) > 2$. Overall, the MMAR has enough flexibility to accommodate a wide variety of tail behaviors.

We can also analyze how the unconditional distribution of returns varies with the time horizon t . Consider for instance a conservative measure μ , such as a random binomial. At the final instant T , the trading time $\theta(T)$ is deterministic, implying that the random variable $X(T)$ is normally distributed. As we move to a smaller horizon t , the allocation of mass becomes increasingly heterogeneous, as is apparent in figure 1. The tails of returns thus become thicker at higher frequencies. The mass of a dyadic cell can be written as $\mu[t, t + 2^{-k}] = m_0^{k\varphi_0} m_1^{k(1-\varphi_0)}$, where $t = 0 \cdot \eta_1 \cdots \eta_k$ and φ_0 denotes the proportion of the

¹² We also know that the scaling function $\tau_\theta(q)$ is negative when $0 < q < 1$ and positive when $1 < q < q_{\text{crit}}(\theta)$.

FIGURE 2.—MMAR SIMULATION WITH RANDOM BINOMIAL TRADING TIME



This figure shows the first differences of a simulation obtained by compounding a standard Brownian motion with a binomial trading time. The simulation displays volatility clustering at all time scales and intermittent large fluctuations.

multipliers $M_{\eta_1}, \dots, M_{\eta_1, \dots, \eta_k}$ equal to m_0 . By the law of large numbers, draws of φ_0 concentrate increasingly in the neighborhood of $1/2$ as k increases, implying that the bell of the distribution becomes thicker. The distribution of $X(t)$ thus accumulates more mass in the tails and in the bell as the time horizon decreases, while the middle of the distribution becomes thinner. This property, which is consistent with empirical observations, is further elaborated by Calvet and Fisher (forthcoming). In addition, when the measure μ is canonical, the random variables $\theta(T)$ and $X(T)$ have Paretian tails, thus illustrating that multifractal returns need not converge to a Gaussian at low frequency.

The model also has an appealing autocorrelation structure.

Theorem 2: The price $\{P(t)\}$ is a semi-martingale (with respect to its natural filtration), and the process $\{X(t)\}$ is a martingale with finite variance and thus uncorrelated increments.

The model thus implies that asset returns have a white spectrum, a property that has been extensively discussed in the market efficiency literature.¹³

The price $P(t)$ is a semi-martingale,¹⁴ which has important consequences for arbitrage.¹⁵ Consider, for instance, the two-asset economy consisting of the multifractal security with price $P(t)$ and a riskless bond with constant rate of return r . Following Harrison and Kreps (1979), we can analyze if arbitrage

profits can be made by frequently rebalancing a portfolio of these two securities. Theorem 2 directly implies

Theorem 3: There are no arbitrage opportunities in the two-asset economy.

This suggests that future research may seek to embed the MMAR in standard financial models. Because the price $P(t)$ is a semi-martingale, stochastic integration can be used to calculate the gains from trading multifractal assets, which in future work will greatly help to develop portfolio selection and option pricing applications. Further research will also seek to integrate multifractality into equilibrium theory. We may thus obtain the MMAR in a general equilibrium model with exogenous multifractal technological shocks, in the spirit of Cox, Ingersoll, and Ross (1985). Such a methodology is justified by the multifractality of many natural phenomena, such as weather patterns, and will help build new economic models of asset and commodity prices. Another line of research could also obtain multifractality as an endogenous equilibrium property, which might stem from the incompleteness of financial markets (Calvet, 2001) or informational cascades (Gennotte & Leland, 1990; Bikhchandani, Hirshleifer, & Welch, 1992; Jacklin, Kleidon, & Pfleiderer, 1992; Bulow & Klemperer, 1994; Avery & Zemsky, 1998).

Recent research focuses not only on predictability in returns, but also on persistence in the size of price changes. The MMAR adds to this literature by proposing a continuous-time model with long memory in volatility. Because the price process is defined only on a bounded time range, the definition of long memory seems problematic. We note, however, that, for any stochastic process Z with stationary increments $Z(a, \Delta t) \equiv Z(a + \Delta t) - Z(t)$, the autocovariance in levels

$$\delta_Z(t, q) = \text{Cov}(|Z(a, \Delta t)|^q, |Z(a + t, \Delta t)|^q),$$

¹³ See Campbell, Lo, and MacKinlay (1997) for a recent discussion of these concepts. We also note that immediate extensions of the MMAR could add trends or other predictable components to the compound process in order to fit different financial time series.

¹⁴ Because the process $X(t) = \ln P(t)$ is a martingale, Jensen's inequality implies that the price $P(t)$ is a submartingale but not a martingale. This result is of course not specific to the MMAR.

¹⁵ See Dothan (1990) for a discussion of semi-martingales in the context of finance.

TABLE 1.—COMPARISON WITH OTHER MODELS

Models		Properties
Volatility Clustering Martingale Returns	Correlated Returns	
MMAR	FBM	Long memory Continuous time
FIGARCH	ARFIMA	Long memory Discrete time
GARCH	ARMA	Short memory Discrete time

quantifies the dependence in the size of the process's increments. It is well defined when $\mathbb{E}|Z(a, \Delta t)|^{2q}$ is finite. For a fixed q , we say that the process has long memory in the size of increments if the autocovariance in levels is hyperbolic in t when $t/\Delta t \rightarrow \infty$. When the process Z is multifractal, this concept does not depend on the particular choice of q .¹⁶ It is easy to show that, when μ is a multiplicative measure,

Theorem 4: Trading time $\theta(t)$ and log-price $X(t)$ have long memory in the size of increments.

This result can be illustrated graphically. Figure 2 shows simulated first differences when $\theta(t)$ is the c.d.f. of a randomized binomial measure with parameter $m_0 = 0.6$. The simulated returns displays marked temporal heterogeneity at all time scales and intermittent large fluctuations.

The MMAR is thus a flexible continuous-time framework that accommodates long memory in volatility, a variety of tail behaviors, and unpredictability in returns. Furthermore, the multifractal model contains volatility persistence at all time frequencies. Table 1 compares the MMAR with existing models of financial time series.

IV. The Multifractal Spectrum

This section examines the geometric properties of sample paths in the MMAR. Although we previously focused on global properties such as moments and autocovariances, we now adopt a more local viewpoint and examine the regularity of realized paths around a given instant. The analysis builds on a concept borrowed from real analysis, the local Hölder exponent. On a given path, the infinitesimal variation in price around a date t is heuristically of the form¹⁷

$$|\ln P(t + dt) - \ln P(t)| \sim C_t(dt)^{\alpha(t)},$$

where $\alpha(t)$ and C_t are respectively called the *local Hölder exponent* and the *prefactor* at t . As is apparent in this definition, the exponent $\alpha(t)$ quantifies the scaling properties of the process at a given point in time, and is also called the *local scale* of the process at t .

In continuous Itô diffusions, the Hölder exponent takes the unique value $\alpha(t) = 1/2$ at every instant.¹⁸ For this reason, traditional research obtains time variations in market volatility through changes in the prefactor C_t . In contrast, the MMAR contains a continuum of local scales $\alpha(t)$ within any finite time interval. Thus, multifractal processes are not continuous Itô diffusions and cannot be generated by standard techniques. Fractal geometry imposes that, in the MMAR, the instants $\{t : \alpha(t) < \alpha\}$ with local scale less than α cluster in clock time, thus accounting for the concentration of outliers in our model. The relative frequency of the local exponents can be represented by a renormalized density called the *multifractal spectrum*. For a broad class of multifractals, we calculate this spectrum by an application of large-deviation theory.

A. Local Scales

We first introduce

Definition 5: Let g be a function defined on the neighborhood of a given date t . The number

$$\begin{aligned} \alpha(t) &= \text{Sup}\{\beta \geq 0 : |g(t + \Delta t) - g(t)| \\ &= O(|\Delta t|^\beta) \quad \text{as } \Delta t \rightarrow 0\} \end{aligned}$$

is called the *local Hölder exponent* or *local scale* of g at t . The Hölder exponent thus describes the local scaling of a path at a point in time, and lower values correspond to more-abrupt variations. The exponent $\alpha(t)$ is nonnegative when the function g is bounded around t , as is always the case in this paper. The definition readily extends to measures on the real line. At a given date t , a measure simply has the local exponent of its c.d.f.

We can easily compute Hölder exponents for many functions and processes. For instance, the local scale of a function is 0 at points of discontinuity, and 1 at (nonsingular) differentiable points. Smooth functions thus have integral exponents almost everywhere. On the other hand, the unique scale $\alpha(t) = 1/2$ is observed on the jagged sample paths of a Brownian motion or of a continuous Itô diffusion. Similarly, a fractional Brownian $B_H(t)$ is characterized by a unique exponent, $\alpha(t) = H$. Thus, the continuous processes typically used in finance each have a unique Hölder exponent. In contrast, multifractal processes contain a continuum of local scales.

The mathematics literature has developed a convenient representation for the distribution of Hölder exponents in a multifractal. This representation, called the *multifractal spectrum*, is a function $f(\alpha)$ that we now describe. From

¹⁶ Provided that $\mathbb{E}|Z(a, \Delta t)|^{2q} < \infty$, as is implicitly assumed in the rest of the paper.

¹⁷ The expression $(dt)^{\alpha(t)}$ is an example of “nonstandard infinitesimal,” as developed by Abraham Robinson.

¹⁸ More precisely, the set $\{t : \alpha(t) \neq 1/2\}$ of instants with a local scale different from $1/2$ has a Hausdorff-Besicovitch measure (and therefore a Lebesgue measure) equal to zero. This set can thus be neglected in our analysis. See Kahane (1997) for a recent survey of this topic.

Definition 5, the Hölder exponent $\alpha(t)$ is the liminf of the ratio

$$\ln |g(t, \Delta t)| / \ln (\Delta t) \quad \text{as} \quad \Delta t \rightarrow 0,$$

where, consistent with previous notation, $g(t, \Delta t) \equiv g(t + \Delta t) - g(t)$. This suggests estimating the distribution of the local scale $\alpha(t)$ at a random instant. For increasing $k \geq 1$, we partition $[0, T]$ into b^k subintervals $[t_i, t_i + \Delta t]$, where length $\Delta t = b^{-k}T$, and calculate for each subinterval the coarse Hölder exponent

$$\alpha_k(t_i) \equiv \ln |g(t_i, \Delta t)| / \ln \Delta t.$$

This operation generates a set $\{\alpha_k(t_i)\}$ of b^k observations. We then divide the range of α 's into small intervals of length $\Delta\alpha$, and denote by $N_k(\alpha)$ the number of coarse exponents contained in $(\alpha, \alpha + \Delta\alpha]$. It would then be natural to calculate a histogram with the relative frequencies $N_k(\alpha)/b^k$, which converge as $k \rightarrow \infty$ to the probability that a random instant t has Hölder exponent α . Using this method, however, the histogram would degenerate into a spike and thus fail to distinguish the MMAR from traditional processes. This is because multifractals typically have a dominant exponent α_0 , in the sense that $\alpha(t) = \alpha_0$ at almost every instant. Mandelbrot (1974, 1989a) instead suggested

Definition 6: The limit

$$f(\alpha) \equiv \lim \left\{ \frac{\ln N_k(\alpha)}{\ln b^k} \right\} \text{ as } k \rightarrow \infty \quad (6)$$

represents a renormalized probability distribution of local Hölder exponents, and is called the *multifractal spectrum*.

For instance, if $b = 3$ and $N_k(\alpha) = 2^k$, the frequency $N_k(\alpha)/b^k = (2/3)^k$ converges to zero as $k \rightarrow \infty$, while the ratio $\ln N_k(\alpha)/\ln b^k = \ln 2/\ln 3$ is a positive constant. The multifractal spectrum thus helps to identify events that happen many times in the construction but at a vanishing frequency.

Frisch and Parisi (1985) and Halsey et al. (1986) interpreted $f(\alpha)$ as the fractal dimension of $T(\alpha) = \{t \in [0, T] : \alpha(t) = \alpha\}$, the set of instants having local Hölder exponent α . For various levels of the scale α , Figure 1d illustrates the subintervals with coarse exponent $\alpha_k(t_i) < \alpha$. When the number of iterations k is sufficiently large, these “cuts” display a self-similar structure. A more detailed discussion of this interpretation can be found in the appendix (subsection F).

B. The Spectrum of Multiplicative Measures

We now use large-deviation theory to compute the spectrum of multiplicative measures. First consider a conservative measure μ defined on the unit interval $[0, 1]$. After k iterations, we know the masses $\mu[t, t + \Delta t] = M_{\eta_1} M_{\eta_2} \cdots M_{\eta_1, \dots, \eta_k}$ in intervals of length $\Delta t = b^{-k}$. The

coarse exponents $\alpha_k(t) = \ln \mu[t, t + \Delta t] / \ln \Delta t$ can thus be rewritten as

$$\alpha_k(t) = -(\log_b M_{\eta_1} + \cdots + \log_b M_{\eta_1, \dots, \eta_k})/k. \quad (7)$$

The multifractal spectrum is obtained by forming renormalized histograms. Letting $V_i \equiv -\log_b M_{\eta_1, \dots, \eta_i}$, we can interpret the coarse Hölder exponents as draws of the random variable

$$\alpha_k = \frac{1}{k} \sum_{i=1}^k V_i. \quad (8)$$

The spectrum $f(\alpha)$ then directly depends on the asymptotic distribution of α_k .

By the Strong Law of Large Numbers, α_k converges almost surely to¹⁹

$$\alpha_0 = \mathbb{E} V_1 = -\mathbb{E} \log_b M > 1. \quad (9)$$

As $k \rightarrow \infty$, almost all coarse exponents are contained in a small neighborhood of α_0 . The standard histogram $N_k(\alpha)/b^k$ thus collapses to a spike at α_0 as anticipated in subsection IVA. The other coarse exponents nonetheless matter greatly. In fact, most of the mass concentrates on intervals with Hölder exponents that are bounded away from α_0 .²⁰ Information on these “rare events” is presumably contained in the tail of the random variable α_k .

Tail behavior is the object of large-deviation theory. In 1938, Cramér established the following theorem under conditions that were gradually weakened.

Theorem 5: Let $\{X_k\}$ denote a sequence of i.i.d. random variables. Then, as $k \rightarrow \infty$,

$$\frac{1}{k} \ln \mathbb{P} \left\{ \frac{1}{k} \sum_{i=1}^k X_i > \alpha \right\} \rightarrow \inf_q \ln [\mathbb{E} e^{q(\alpha - X_1)}],$$

for any $\alpha > \mathbb{E} X_1$.

Proofs can be found in Billingsley (1979) and Durrett (1991). The theorem implies

Theorem 6: The multifractal spectrum $f(\alpha)$ is the Legendre transform

$$f(\alpha) = \inf_q [\alpha q - \tau(q)] \quad (10)$$

of the scaling function $\tau(q)$.

¹⁹ The relation $-\mathbb{E} \log_b M > 1$ follows from Jensen's inequality and $\mathbb{E} M = 1/b$.

²⁰ Let T_k denote the set of b -adic cells with local exponents greater than $(1 + \alpha_0)/2$. When k is large, T_k contains “almost all” cells, but its mass

$$\sum_{t \in T_k} (\Delta t)^{\alpha_k(t)} \leq b^k (\Delta t)^{(\alpha_0+1)/2} = b^{-k(\alpha_0-1)/2}$$

vanishes as k goes to infinity.

TABLE 2.—EXAMPLES OF MULTIFRACTAL SPECTRA

Distribution of V	Multifractal Spectrum $f_\theta(\alpha)$
Normal (λ, σ^2)	$1 - (\alpha - \lambda)^2/[4(\lambda - 1)]$
Binomial	$-\frac{\alpha_{\max} - \alpha}{\alpha_{\max} - \alpha_{\min}} \log_2 \left(\frac{\alpha_{\max} - \alpha}{\alpha_{\max} - \alpha_{\min}} \right) - \frac{\alpha - \alpha_{\min}}{\alpha_{\max} - \alpha_{\min}} \log_2 \left(\frac{\alpha - \alpha_{\min}}{\alpha_{\max} - \alpha_{\min}} \right)$
Poisson (γ)	$1 - \gamma/\ln b + \alpha \log_b (\gamma e/\alpha)$
Gamma (β, γ)	$1 + \gamma \log_b (\alpha \beta/\gamma) + (\gamma - \alpha \beta)/\ln b$

This table shows the multifractal spectrum of a multiplicative measure and its corresponding trading time when the random variable $V = -\log_b M$ is respectively: (1) a Gaussian density of mean λ and variance σ^2 , (2) a binomial distribution taking discrete values $\alpha_{\min}$ and $\alpha_{\max}$ with equal probability, (3) a discrete Poisson distribution $p(x) = e^{-\gamma} \gamma^x / x!$, and (4) a Gamma distribution with density $p(x) = \beta^\gamma x^{\gamma-1} e^{-\beta x} / \Gamma(\gamma)$.

This result holds for both conservative and canonical measures. It provides the foundation of the empirical work developed in section VI, in which an estimation procedure for the scaling function $\tau(q)$ is obtained and the Legendre transform yields an estimate of the multifractal spectrum $f(\alpha)$.

The theorem allows us to derive explicit formulas for the spectrum in a number of useful examples. To aid future reference, we denote by $f_\theta(\alpha)$ the spectrum common to a measure μ and its c.d.f. θ . Begin by considering a measure generated by a log normal multiplier M with distribution $-\log_b M \sim \mathcal{N}(\lambda, \sigma^2)$. Conservation of mass imposes that $\mathbb{E}M = 1/b$, or, equivalently, $\sigma^2 = 2 \ln b/(\lambda - 1)$. It is easy to show that the scaling function $\tau(q) \equiv -\log_b (\mathbb{E}M^q) - 1$ has the closed-form expression $\tau(q) = \lambda q - 1 - q^2 \sigma^2 (\ln b)/2$. We infer from Theorem 6 that the multifractal spectrum is a quadratic function

$$f_\theta(\alpha) = 1 - (\alpha - \lambda)^2/[4(\lambda - 1)]$$

parameterized by a unique number, $\lambda > 1$. More generally, table 2 reports the spectrum when the random variable V is binomial, Poisson, or Gamma. (See CFM (1997) for detailed derivations.) We note that the function $f_\theta(\alpha)$ is very sensitive to the distribution of the multiplier, which suggests that the MMAR has enough flexibility to model a wide range of financial prices. In the empirical work, this allows us to identify a multiplicative measure from its estimated spectrum.

C. Application to the MMAR

We now examine the spectrum of price processes generated by the MMAR. Denoting by $f_Z(\alpha)$ the spectrum of a process $Z(t)$, we show

Theorem 7: The price $P(t)$ and the log-price $X(t)$ have identical multifractal spectra: $f_P(\alpha) \equiv f_X(\alpha) \equiv f_\theta(2\alpha)$.

The log-price $X(t)$ contains a continuum of local exponents and thus cannot be generated by an Itô diffusion process. Let $\alpha_0(Z)$ denote the most probable exponent of a process Z . Because $\alpha_0(\theta) > 1$, the log-price has a local scale $\alpha_0(X) \equiv \alpha_0(\theta)/2$ larger than 1/2 at almost every instant. Despite their apparent irregularity, the MMAR's sample paths are almost everywhere smoother than the paths of a

Brownian motion. Subsection IVB indicates that the variability of the MMAR is in fact explained by the "rare" local scales $\alpha < \alpha_0(X)$. Although jump diffusions permit negligible sets to contribute to the total variation, multifractal processes are notable for combining continuous paths with variations dominated by rare events.

Although the local scale is larger than 1/2 almost everywhere, Theorem 1 implies that the standard deviation of the process

$$\sqrt{\mathbb{E}\{[X(t + \Delta t) - X(t)]^2\}} = c_X(2)^{1/2} \sqrt{\Delta t}$$

is of the order $(\Delta t)^{1/2}$. Thus, whereas most shocks are of order $(\Delta t)^{\alpha_0(X)}$, the exponents $\alpha < \alpha_0(X)$ appear frequently enough to alter the scaling properties of the variance. This contrasts with the textbook analysis that a standard deviation in $(\Delta t)^{1/2}$ implies that most shocks are of the same order.²¹ We expect these findings to have interesting consequences for decision and equilibrium theory.

V. An Extension with Autocorrelated Returns

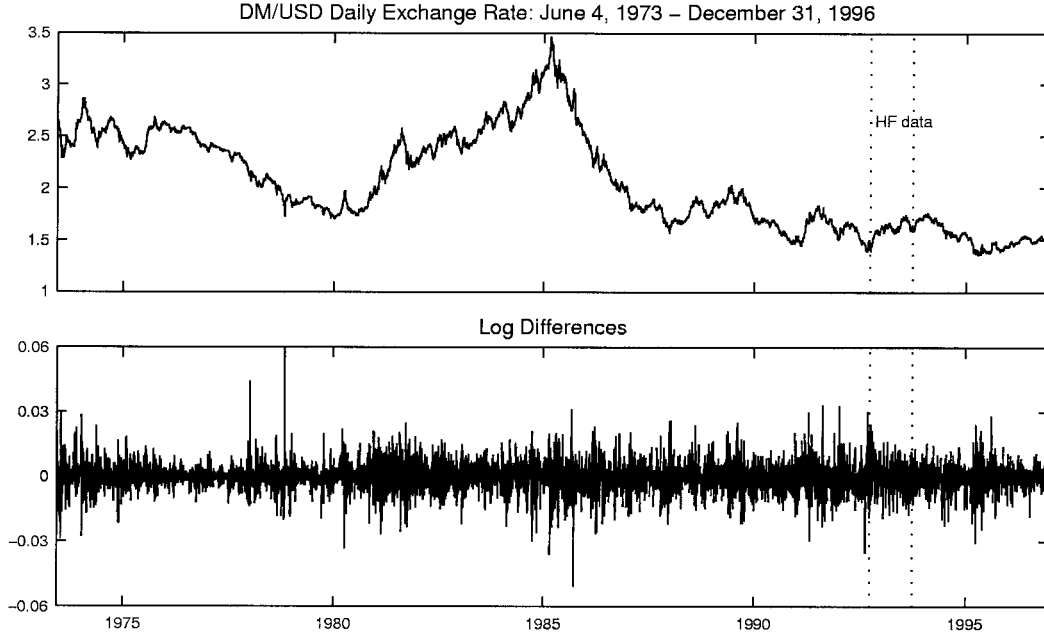
The multifractal model presented in section III is characterized by long memory in volatility but the absence of correlation in returns. Although there is little evidence of fractional integration in stock returns (Lo, 1991), long memory has been identified in the first differences of many economic series,²² including aggregate output (Adelman, 1965; Diebold & Rudebusch, 1989; Sowell, 1992), the Beveridge (1921) wheat price index, the U.S. consumer price index (Baillie, Chung, & Tieslau, 1996), and interest rates (Backus & Zin, 1993).²³ This has led authors to model these series with the FBM or the discrete-time ARFIMA

²¹ Merton (1990, ch. 3) provides an interesting discussion of multiple local scales and "rare events" in financial processes. Assume that the price variation over a time interval Δt is a discrete random variable taking values $\epsilon_1, \dots, \epsilon_m$ with probability $p_1, \dots, p_m$, and assume moreover that $p_i \sim (\Delta t)^{q_i}$, $\epsilon_i \sim (\Delta t)^{r_i}$ and $r_i > 0$ for all i . Denote by I the events i such that $p_i \epsilon_i^2 \sim \Delta t$. When the variance of the process $\sum_{i=1}^m p_i \epsilon_i^2$ is of the order Δt , only events in I contribute to the variance. If all events belong to I , Merton establishes that only events of the order $(\Delta t)^{1/2}$ matter. The MMAR shows that events outside I can play a crucial role in the statistical properties of the price process, a property previously overlooked in the literature.

²² Maheswaran and Sims (1993) suggest potential applications in finance for processes lying outside the class of semi-martingales.

²³ See Baillie (1996) for a review of this literature.

FIGURE 3.—DM/USD DAILY DATA



The data is provided by Olsen and Associates and spans from June 1973 to December 1996. The outlined area labeled “HF data” shows the one-year period from October 1992 to October 1993 that corresponds to the span of our high-frequency data. Despite the apparent long cycles and clustering of volatility in the daily data, the MMAR provides sufficient flexibility to capture this behavior in a parsimonious stationary model. Moreover, both the daily data and the high-frequency data show similar scaling patterns.

specification. We note, however, that these economic series have volatility patterns that seem closer to the multifractal model than to the fractional Brownian motion. This suggests using the fractional Brownian motion of multifractal time.

We model an economic series $X(t)$ by replacing Assumption 1 in section III with

Assumption 1a: $X(t)$ is a compound process $X(t) \equiv B_H[\theta(t)]$ where $B_H(t)$ is a fractional Brownian motion, and $\theta(t)$ is a stochastic trading time.

In addition, we maintain the multifractality of trading time (Assumption 2) and the independence of the processes $B_H(t)$ and $\theta(t)$ (Assumption 3). Note that this coincides with the earlier model if $H = 1/2$. For other values of the index H , the increments of $X(t)$ display either antipersistent ($H < 1/2$) or positive autocorrelations and long memory ($H > 1/2$). The more general model is fully developed by MFC, CFM, and FCM (1997).

The self-similarity of $B_H(t)$ implies

Theorem 8: The process $X(t)$ is a multifractal process with stationary increments, scaling function $\tau_X(q) = \tau_\theta(Hq)$, and multifractal spectrum $f_X(\alpha) = f_\theta(\alpha/H)$.

The proof of these results is provided by MFC (1997). We observe that $\tau_X(1/H) = \tau_\theta(1) = 0$, which allows the estimation of the index H in the empirical work. The generalized construction has scaling properties analogous to the model explored earlier, and provides a useful additional tool for empirical applications.

VI. Empirical Evidence

A. Multifractal Moment Restrictions

Consider a price series $P(t)$ on the time interval $[0, T]$, and the log-price $X(t) \equiv \ln P(t) - \ln P(0)$. Partitioning $[0, T]$ into integer N intervals of length Δt , we define the sample sum or partition function

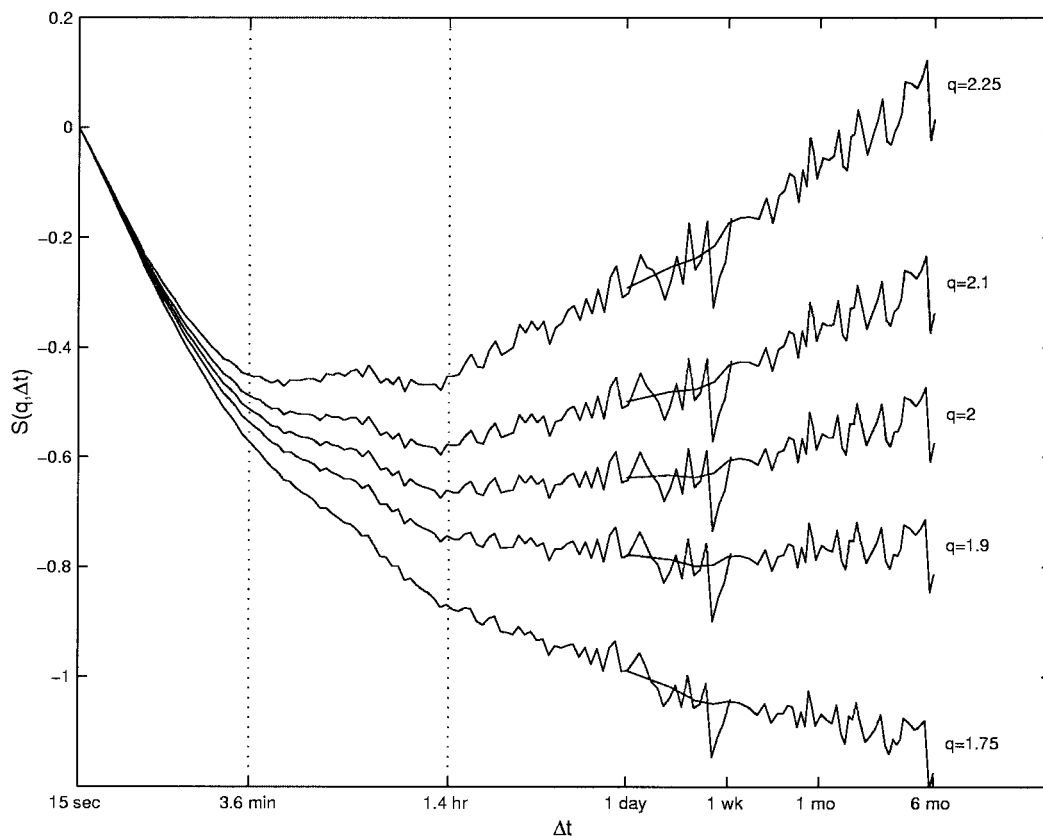
$$S_q(T, \Delta t) \equiv \sum_{i=0}^{N-1} |X(i\Delta t + \Delta t) - X(i\Delta t)|^q. \quad (11)$$

When $X(t)$ is multifractal, the addends are identically distributed, and the scaling law (5) yields $\mathbb{E}[S_q(T, \Delta t)] = Nc(q)(\Delta t)^{\tau_X(q)+1}$ when the q^{th} moment exists. This implies

$$\ln \mathbb{E}[S_q(T, \Delta t)] = \tau_X(q) \ln(\Delta t) + c^*(q), \quad (12)$$

where $c^*(q) = \ln c(q) + \ln T$. For each admissible q , equation (12) provides testable moment conditions describing how the partition function varies with increment size Δt . Various methods can be used to construct an estimator $\hat{\tau}_X(q)$ from the sample moments of the data. By equation (10), its Legendre transform $\hat{f}(\alpha)$ provides an estimate of the multifractal spectrum, and can then be mapped back into a distribution of the multipliers.

The scaling function $\tau_X(q)$ is specified either parametrically or nonparametrically. We can, for instance, choose a parametric family for the distribution of the multiplier M . The multifractal spectrum $f(\alpha)$ then belongs to a specific

FIGURE 4.—DM/USD PARTITION FUNCTIONS FOR THE FULL SET OF TIMESCALES AND MOMENTS q NEAR 2

This figure identifies a scaling region from about 1.4 hours to at least six months, the largest horizon for which the partition function was calculated. As discussed in subsection VIC, the change in scaling at high frequencies is consistent with market frictions such as bid-ask spread, discreteness of trading units, and noncontinuous trade. Moments q near 2 were chosen to investigate the martingale hypothesis for returns by the equation $\tau(q = 1/H) = 0$. We find a flat slope near $q = 1.88$, implying $\hat{H} = 0.53$, or slight persistence. Using simulation methods, subsection VID investigates whether this evidence is sufficient to reject the martingale version of the MMAR.

class of functions (table 2), a constraint that can be imposed in estimation. On the other hand, a nonparametric approach places fewer restrictions on the underlying process. Because $\tau_\theta(q) = -\log_b \mathbb{E}(M^q) - 1$, the sample moments provide all the finite moments of M , and thus a great deal of information on its underlying distribution.²⁴

This paper uses a very simple estimation procedure. Given a set of positive moments q and time scales Δt , we calculate the partition functions $S_q(T, \Delta t)$ of the data. The partition functions are then plotted against Δt in logarithmic scales. By equation (12), the multifractal model implies that these plots should be approximately linear when the q^{th} moment exists. Regression estimates of the slopes then provide the corresponding scaling exponents, $\hat{\tau}_X(q)$. This procedure reveals striking visual evidence of moment scaling in DM/USD data. Simulation experiments are then conducted to assess the joint performance of the multifractal model and the estimation methodology.

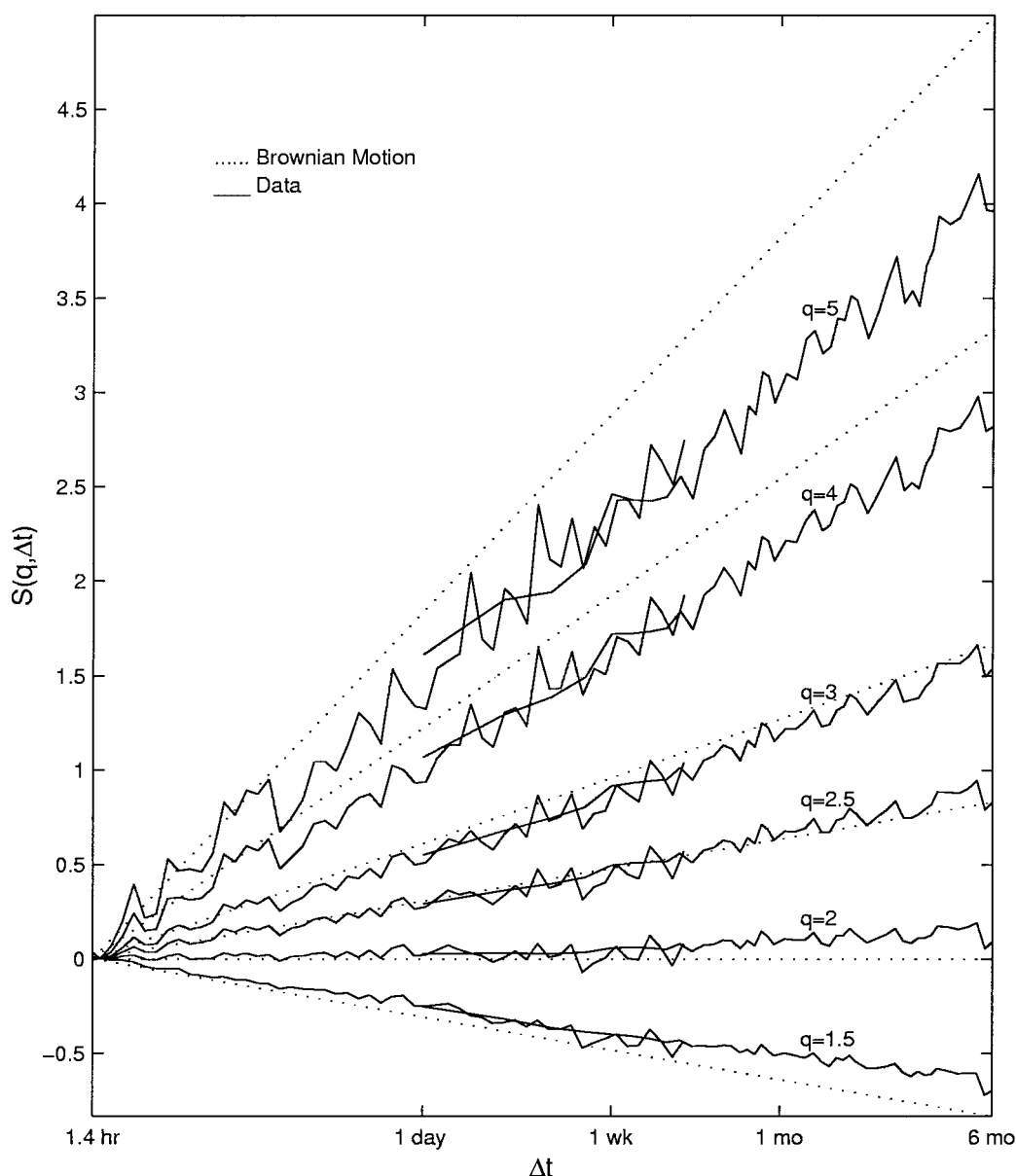
²⁴ The distribution of M may not be uniquely determined by its moments. See Feller (1968) and Durrett (1991) for good discussions of the uniqueness problem in moments.

B. Deutsche Mark/U.S. Dollar Exchange Rates

We begin by investigating the multifractality of the Deutsche mark/U.S. dollar (DM/USD) exchange rate. We use two data sets provided by Olsen and Associates, a currency research and trading firm based in Zürich. The first data set ("daily") consists of a 24-year series of daily data spanning June 1973 to December 1996. Olsen collects price quotes from banks and other institutions through several electronic networks. A price quote is converted to a single price observation by taking the geometric mean of the concurrent bid and ask. The reported price in the daily data is then calculated by linear interpolation of the price observations closest to 16:00 local UK time on each side.²⁵ Figure 3 shows the daily data, which exhibits volatility clustering at all time scales and intermittent large fluctuations.

The second data set ("high-frequency") contains all bid/ask quotes and transmittal times collected over the one-year period from October 31, 1992, to September 1, 1993. We

²⁵ An earlier working paper (FCM, 1997) also uses noon buying rates provided by the Federal Reserve, and finds no significant difference in reported results.

FIGURE 5.—DM/USD PARTITION FUNCTIONS IN THE SCALING REGION FOR MOMENTS $1.5 \leq q \leq 5$ 

For each moment, the first solid line plotted from 1.4 hours to two weeks corresponds to the high-frequency data. The second solid line ranges from $\Delta t = 1$ day to six months, and corresponds to the daily data. The lines are remarkably straight, as predicted by the model, and have nearly identical slopes. Also, their scaling is noticeably different from that of the Brownian motion, which is shown by the dotted lines.

convert quotes to price observations using the same methodology as Olsen, and obtain a round-the-clock data set of 1,472,241 observations. Olsen provides a flag for quotes that are believed to be erroneous or not representative of actual willingness to trade. We eliminate these observations, which constitute 0.36% of the data set. Combining the daily data and the high-frequency data allows us to calculate partition functions over three orders of magnitude for Δt .

The high-frequency data show strong patterns of daily seasonality. In continuous time, seasonality is a smooth transformation that does not affect local Hölder exponents. Because our data is discrete, however, we may expect

seasonality to introduce noise. To reduce this effect, we can write a seasonally modified version of the MMAR:

$$\ln P(t) - \ln P(0) = B_H\{\theta[SEAS(t)]\},$$

where the seasonal transformation $SEAS(t)$ is a differentiable function of clock time. In this paper, we use a prefilter that smoothes variation in average absolute returns over fifteen-minute intervals of the week. Except for the reduction in noise, there are no systematic differences in reported results for filtered and unfiltered data. An earlier working paper (FCM, 1997) provides details on this and

three other seasonal prefilters, and finds small, predictable differences in results depending on the deseasonalizing method.

C. Main Results

Figures 4 and 5 illustrate the partition functions of the two DM/USD data sets. Values of Δt are chosen to increase multiplicatively by a factor of 1.1 from minimum to maximum. Because we focus on the slopes $\hat{\tau}_X(q)$ but not the intercepts, plots for each q are renormalized by vertical displacement to begin at zero for the lowest value of Δt in each graph. This allows plots for many q to be presented simultaneously. The daily and high-frequency plots are presented in the same graph to highlight the similarity in their slopes. This is achieved by a second vertical displacement of the daily data that provides the best linear fit under OLS, restricting both lines to have the same slope.

Figure 4 shows the full range of calculated Δt , from fifteen seconds to six months, and five values of q ranging from 1.75 to 2.25. This allows us to estimate the self-affinity index H in the extended model presented in section V. Because $\tau_X(1/H) = 0$ and the standard Brownian specification $H = 1/2$ has previous empirical support, we expect to find $\tau_X(q) = 0$ for a value of q near 2.

We first note the approximate linearity of the partition functions beginning at $\Delta t = 1.4$ hours and extending to the largest increment used, $\Delta t = 6$ months. In this range, the slope is zero for a value of q slightly smaller than 2, and we report $\hat{H} \approx 0.53$, which implies very slight persistence in the DM/USD series. It is not immediately clear whether this result is sufficiently close to $H = 1/2$ to be consistent with the martingale version of the MMAR, but we will return to this issue in the following section using simulation methods.

The partition functions in figure 4 also show breaks in linearity at high frequencies. These are consistent with microstructure effects such as bid-ask spreads, discreteness of quoting units, and discontinuous trading. In particular, these microstructure effects can be expected to induce a negative autocorrelation at high frequencies, as is well understood in the case of bid-ask bounce (Roll, 1984). Negative autocorrelation effectively acts as an additional source of volatility, as previously explored in the variance ratio literature (Campbell & Mankiw, 1987; Lo & MacKinlay, 1988; Richardson & Stock, 1989; Faust, 1992). The results in figure 4 are analogous to variance ratio tests, exactly so if we focus on the moment $q = 2$. As we move to the left on the graph and sampling frequency increases, microstructure-induced negative autocorrelation increases, and the plots bend upwards corresponding to the increase in variability.

Descriptive statistics help to confirm that high-frequency breaks in linearity are caused by microstructure effects. The departure from linearity begins at a frequency of approximately $\Delta t = 1.4$ hours, which is highlighted by the dotted line in figure 4. We first note that the absolute change in the

DM/USD rate averages 0.14 pfennig²⁶ over a time increment of 1.4 hours. Comparing this to the average spread of 0.07 pfennig,²⁷ we observe that the spread covers a significant proportion of average variation at this time horizon. It is thus sensible that microstructure effects begin to effect scaling properties at this frequency. To further confirm this intuition, observe that for time scales between 3.6 minutes and 1.4 hours, the partition function has an approximate slope of zero for the moment $q = 2.25$. This implies that $H \approx 0.44 < 1/2$, which is consistent with the explanation that microstructure-induced negative autocorrelation causes the observed departure from scaling in this high-frequency region.²⁸

These high-frequency microstructure effects have several potential solutions. First, we could view the MMAR as an underlying price process and overlay it with a structural model of discrete trade and bid-ask spreads that could then be related to the observed Olsen quotes. This would permit using the data at all frequencies, but at the cost of introducing a new set of modeling issues. A second alternative consists of prefiltering the data to remove high-frequency autocorrelations. This approach has recently been employed by Andersen, Bollerslev, Diebold, and Ebens (2001), who use an MA(1) filter to remove first-order serial correlation from five-minute returns on the thirty stocks tracked in the Dow Jones Industrial Index. Although this would certainly remove high-frequency autocorrelations, the effect of this procedure on scaling properties would require further investigation. For simplicity, we choose to discard from further analysis all values of Δt less than 1.4 hours,²⁹ and have three orders of magnitude of sampling frequencies with which to test the scaling properties of the data.

With attention now restricted to values of Δt between 1.4 hours and six months, figure 5 presents partition functions for a larger range of moments: $1.5 \leq q \leq 5$. Higher moments capture information in the tails of the distribution of returns, and are thus generally more sensitive to deviations from scaling. All of the plots are nonetheless remarkably linear, and the overlapping values from the two data sets appear to have almost the same slope. Thus, despite the apparent nonstationarity of the 24-year series, such as long price swings and long cycles of volatility, the moment

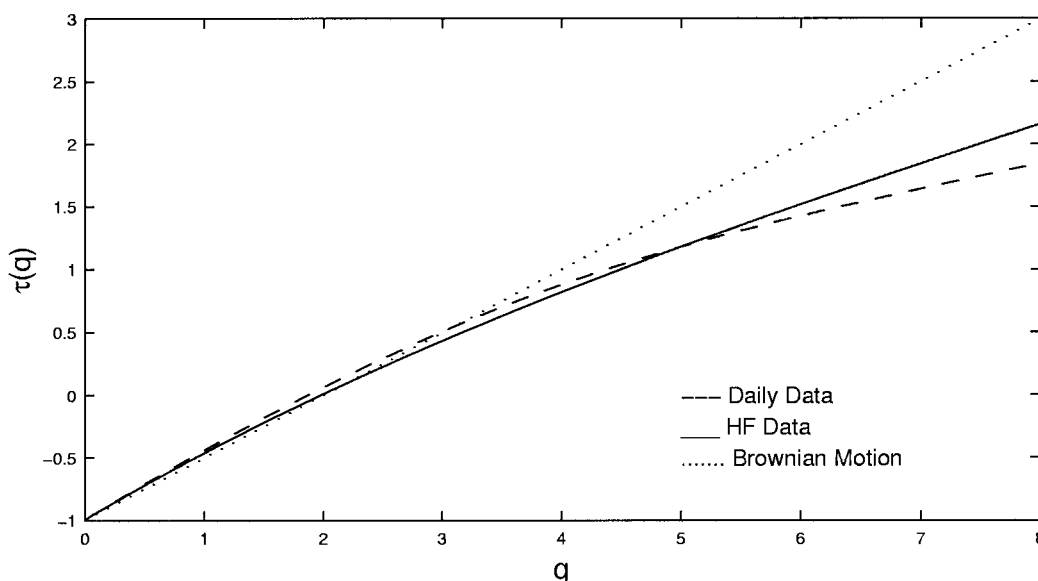
²⁶ One pfennig equals 0.01 DM.

²⁷ The two most common spread sizes are 0.05 pfennig (38.25%), and 0.10 pfennig (52.55%), together comprising more than 90% of all observed spreads.

²⁸ To further confirm this explanation, one could compare across assets the frequency at which departures from moment scaling begin. If the microstructure explanation is correct, one would expect scaling to extend to higher frequencies when trading frictions (measured by variables such as bid-ask spreads or the average time interval between trades) are lower.

²⁹ As an approximation, the choice of $\Delta t = 1.4$ hours as the high-frequency cutoff for our analysis is justified by figure 4, but the exact value was chosen by ad hoc rounding. In the future, there may be some gains to adopting a more formal test for the high-frequency cutoff value. There is already a substantial econometric literature (such as Andrews (1993)) that could be adapted to this purpose.

FIGURE 6.—ESTIMATED DM/USD SCALING FUNCTIONS



For each partition function $S_q(T, \Delta t)$, we estimate the slope using OLS to obtain $\hat{\tau}_X(q)$. The estimated scaling functions for both data sets are concave and have a similar shape until high moments are reached. The contrast with Brownian motion is shown by the dotted line.

restrictions imposed by the MMAR seem to hold over a broad range of sampling frequencies.

Estimates of the slopes in figure 5 and additional moments q are then used to obtain estimated scaling functions $\hat{\tau}_X(q)$ for both data sets. We note the increasing variability of the partition function plots with the time scale Δt , which can be attributed to the shrinking number of addends in the partition function at low frequencies. This suggests a weighted least squares or generalized least squares approach. In practice, however, weighting the observations has little effect on the results because the plots are very nearly linear. Preferring simplicity, we thus report in figure 6 the estimated scaling functions from OLS regressions. The estimated scaling functions are strictly concave, indicating multifractality, and are fairly similar except for very large moments.

Theorem 6 suggests to estimate the multifractal spectrum $f_X(\alpha)$ by taking the Legendre transform of $\hat{\tau}_X(q)$. Following this logic, figure 7 shows the estimated multifractal spectrum of the daily data.³⁰ The estimated spectrum is concave, in contrast to the degenerate spectra of Brownian motion and other unifractals. Using the estimated spectrum, we can recover a generating mechanism for trading time based on the canonical multiplicative cascades described in subsection IIB.

The spectrum of daily data is very nearly quadratic, and subsection IVB has shown that quadratic spectra are generated by log normally distributed multipliers M . We thus specify $-\log_b M \sim \mathcal{N}(\lambda, \sigma^2)$, giving trading time $\theta(t)$ with multifractal spectrum $f_\theta(\alpha) = 1 - (\alpha - \lambda)^2/[4(\lambda - 1)]$.

³⁰ The estimated multifractal spectrum of the high-frequency data is similar in many respects, and is discussed by FCM (1997).

The log-price process has most probable exponent $\alpha_0 = \lambda H$, and spectrum

$$f_X(\alpha) = 1 - \frac{(\alpha - \alpha_0)^2}{4H(\alpha_0 - H)}$$

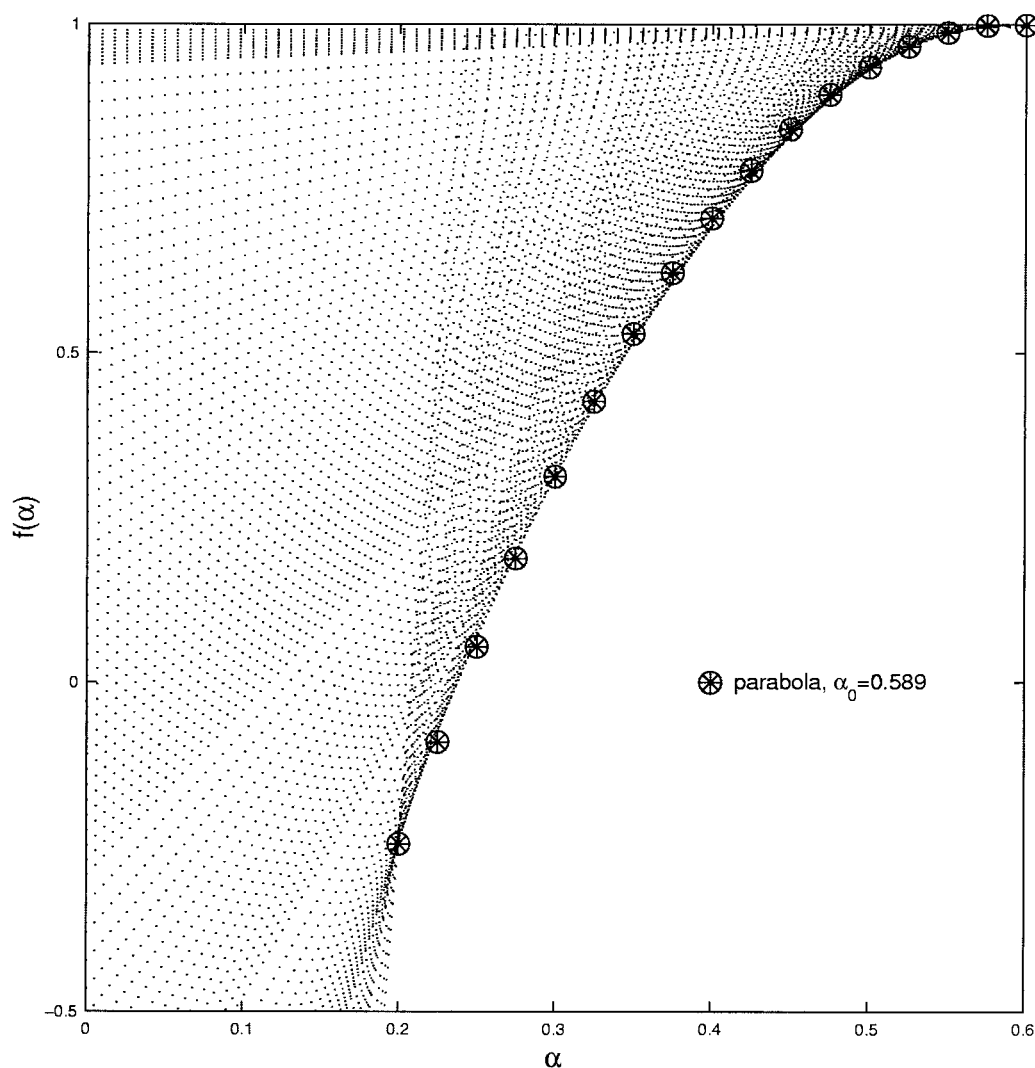
Because $\hat{H} = 0.53$, the free parameter α_0 is used to fit the estimated spectrum. We report $\hat{\alpha}_0 = 0.589$, which produces the parabola shown in figure 7. Choosing a generating construction with base $b = 2$,³¹ this immediately implies that $\hat{\lambda} = 1.11$ and $\hat{\sigma}^2 = 0.32$.³² It is also natural to consider the martingale version of the MMAR with the restriction $H = 1/2$. For this case, we estimate the single parameter $\hat{\alpha}_0 = 0.545$.

In both cases, the estimated value of the most probable local Hölder exponent α_0 is greater than $1/2$. On a set of Lebesgue measure 1, the estimated multifractal process is therefore more regular than a Brownian motion. However, the concavity of the spectrum also implies the existence of lower Hölder exponents that correspond to more-irregular instants of the price process. These contribute disproportionately to volatility.

³¹ The base b of the multifractal generating process is not uniquely identified by the spectrum alone; hence, we assume the commonly used value $b = 2$. Calvet and Fisher (2001) develop a likelihood-based filter under which b can be estimated for the class of multinomial multifractals.

³² Mandelbrot (1989a, 1989b) shows that the partition function methodology provides reasonable estimates of $\tau_X(q)$ only for moments $q < 1/\sqrt{\alpha_0(X)/H - 1}$, which is approximately equal to 5.66 in our estimated process.

FIGURE 7.—ESTIMATED MULTIFRACTAL SPECTRUM OF DAILY DM/USD DATA



The estimated spectrum is obtained from the Legendre transform $\hat{f}_h(\alpha) = \inf_q [\alpha q - \hat{\tau}(q)]$, shown in this graph by the lower envelope of the dotted lines. The shape is nearly quadratic, with a fitted parabola shown by marked symbols. A quadratic spectrum implies a log normal distribution for multipliers M in the multifractal-generating mechanism.

D. Monte Carlo Simulations

We now present simulation experiments that provide a preliminary assessment of the new model and the estimation procedure. Figure 8 shows the levels and log differences of a random price path generated by the limit log normal MMAR estimated in subsection VIC.³³ The simulation shows a variety of large price changes, apparent trends, persistent bursts of volatility, and other characteristics found in the DM/USD series.

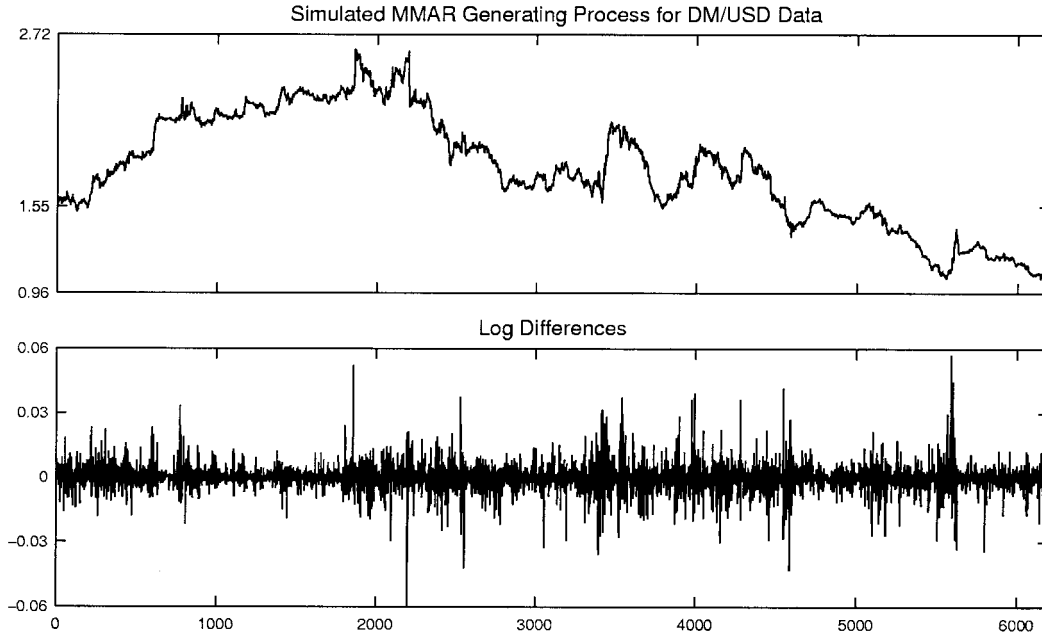
The following sections examine whether the inferred process captures the moment properties of the data. We sketch the simulation methodology, and then provide a synthetic discussion of the numerical results.

³³ The simulation of a multifractal path is discussed in the appendix (subsection I).

Methodology: We use three types of tests to analyze the MMAR's performance. First, visual evidence is provided on the moment properties of simulated data. Figure 9a thus illustrates the partition functions corresponding to four simulations of the MMAR. For comparison, we report in figure 9b the partition functions of a GARCH(1, 1) process with the parameter estimates of Baillie and Bollerslev (1989). Figure 9c similarly considers the FIGARCH (1, d , 0) specification of Baillie, Bollerslev, and Mikkelsen (1996). Each plot in figure 9 is based on a long sample of 100,000 observations. This sample length—which exceeds the sample size of 6,118 daily DM/USD returns—has the advantage of reducing the noisiness of the partition functions.

Second, we consider small samples and examine distributional evidence on the linearity and slopes of the partition functions. The analysis focuses on four processes: the ex-

FIGURE 8.—SIMULATED MULTIFRACTAL-GENERATING PROCESS FOR THE DM/USD DATA



We use the estimated values of $\hat{H} = 0.53$ and $\hat{\alpha}_0 = 0.589$ with the limit log normal construction of trading time. The plots show volatility clustering at all time scales and occasional large fluctuations.

tended MMAR (with arbitrary H), the martingale MMAR ($H = 1/2$), FIGARCH, and GARCH. These models are respectively indexed by $m \in \{1, \dots, 4\}$. For each model m , we simulate $J = 10,000$ paths with the same length $T = 6,118$ as the DM/USD data. We denote each path by $Y_j^m = \{Y_{j,t}^m\}_{t=1}^T$, ($1 \leq j \leq J$), and focus the analysis on the moments $q \in Q = \{0.5, 1, 2, 3, 5\}$. For each path and each q , an OLS regression provides a slope estimate $\hat{\tau}(q, Y_j^m)$ and the corresponding sum of squared errors $SSE(q, Y_j^m)$. The distributions of these statistics appear unimodal with smoothly declining tails. Tables 3 and 4 report the percentiles of these statistics.

We summarize these findings with several measures of global fit. For a given model $m \in \{1, \dots, 4\}$, each path Y_j^m generates a column vector of slope and SSE estimates:³⁴

$$h(Y_j^m) = \{[\hat{\tau}(q, Y_j^m), \ln SSE(q, Y_j^m)]_{q \in Q}\}'.$$

It is convenient to denote the data by $X = \{X_t\}_{t=1}^T$, and to arrange the simulated paths in a $J \times T$ matrix $Y^m = [Y_1^m, \dots, Y_J^m]'$. We also consider

$$H(X, Y^m) = h(X) - \frac{1}{J} \sum_{j=1}^J h(Y_j^m).$$

The function H is useful to test how a particular model fits the moment properties of the data. In particular, we can define a global statistic $G = H'WH$ for any positive-definite matrix W . The empirical work consider four differ-

ent weighting matrices W_m , $m \in \{1, \dots, 4\}$, each of which is obtained by inverting the simulated covariance matrix of moment conditions: $W_m = [\sum_{j=1}^J H(Y_j^m, Y^m)H(Y_j^m, Y^m)'/J]^{-1}$. The global statistics

$$G_{m,n}(X) = H(X, Y^m)' W_n H(X, Y^m), \quad m, n \in \{1, \dots, 4\}.$$

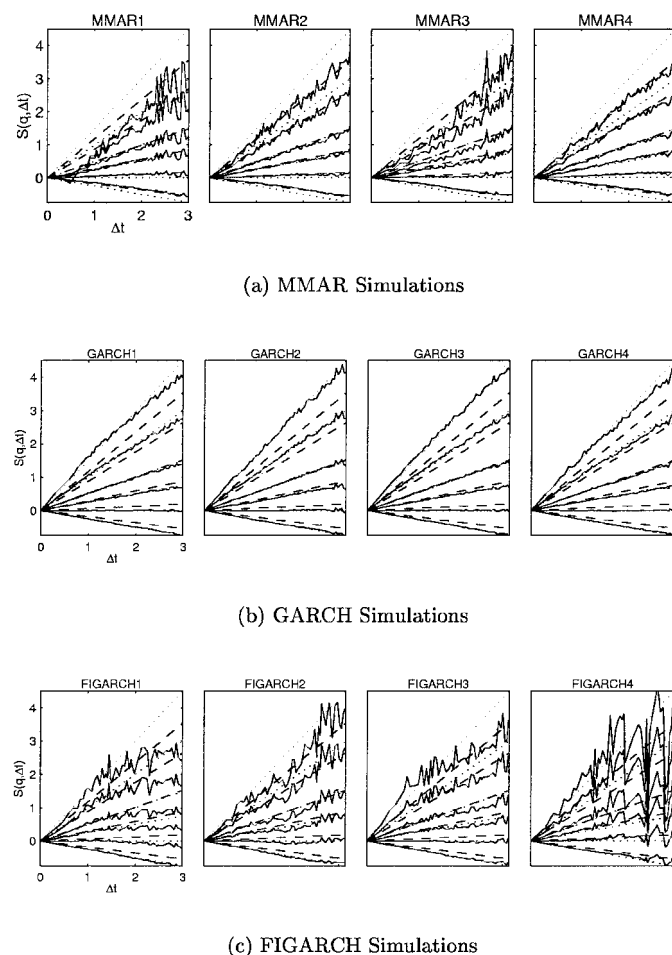
are indexed by the model m that generates the simulated data Y^m and the model n that generates the weighting matrix W_n . This gives a set of sixteen global statistics. Assuming that m is the true model, we can estimate the cumulative distribution function $\mathbb{F}_{m,n}$ of each statistic from the set $\{G_{m,n}(Y_j^m)\}_{1 \leq j \leq J}$, and then quantify the p -value $1 - \mathbb{F}_{m,n}[G_{m,n}(X)]$. The global statistics $G_{m,n}(X)$ and their associated p -values are reported in table 5.

Results: The simulation results in figure 9 and tables 3 through 5 confirm that the MMAR replicates the main scaling features of the data. The partition function plots in figure 9a are approximately linear and tend to follow their theoretically predicted slopes, which are nearly identical to the estimates from the DM/USD data. Tables 3 and 4 permit a more detailed assessment. The extended MMAR is very close to the data in both its theoretically predicted slopes τ_0 and the mean slopes $\bar{\tau}$. Furthermore, the estimated slopes from the DM/USD data are well within the central bells of the simulated slope distributions generated by the extended MMAR.

The martingale version is subtly different in both its theoretically predicted and mean slopes. For low moments,

³⁴ We use the logarithm of the SSE in calculating the global statistics because table 4 shows that the SSE are heavily right-skewed.

FIGURE 9.—SIMULATED PARTITION FUNCTIONS



Each panel shows estimated partition functions for a simulated sample of 100,000 observations. The data-generating process in (a) uses our estimates from the extended MMAR. The data-generating processes in (b) and (c) use specifications from previously published research on daily DM/USD exchange rates. Large simulated samples are used to reduce noise. Dotted lines in each figure represent the scaling predicted for Brownian motion and dashed lines represent the scaling found in the data. The MMAR appears most likely to capture scaling in the DM/USD data. The GARCH simulations tend to scale like Brownian motion. FIGARCH simulations occasionally show scaling that is similar to the data, as in panel (c2), but in general tend to be much more irregular.

the estimated slopes from the DM/USD data are more towards the upper tails of the simulated distributions generated by the martingale MMAR. In both cases, but more so for the martingale version, there appears to be a slight downward bias in the average simulated slope relative to its theoretical value. Future work may thus correct the bias in our estimation method by matching simulated moments (Ingram & Lee, 1991; Duffie & Singleton, 1993).

Table 4 analyzes the variability of the simulated partition functions around their slopes. The extended and martingale versions of the MMAR yield nearly identical results. For low moments, the data falls well within the likely range of the SSE statistic for both models. For high moments, the partition functions are typically more variable for the simulated MMAR than for the data. This at first seems curious because the model has been designed specifically to produce scaling. It is consistent, however, with the finding of a slight downward bias in the slopes of the estimated partition

functions. Correcting this bias would give a slightly milder multifractal process and thus reduce the variability of the partition functions. This presents a promising avenue for improving estimation. Overall, these results suggest that the MMAR is successful in matching the main scaling features of the data.

Another important question is whether other standard econometric models possess scaling properties. Figure 9b shows that GARCH(1, 1) partition functions are fairly linear, but their apparent slope is similar to the predicted slope of Brownian motion rather than the data. This is symptomatic of the fact that GARCH models are short-memory processes. Over long time periods, temporal clustering disappears and GARCH scales like a Brownian motion. Tables 3 and 4 confirm this visual evidence. The SSE statistics show that GARCH tends to be as linear as the data, but, for two of the five moments, the slopes from the data are in the extreme tails of their distributions simulated under GARCH. Because it contains long memory in volatility, FIGARCH can be expected to scale differently than Brownian motion at low frequencies. This is confirmed in figure 9c; however, the same plots suggest that simulated FIGARCH partition functions are more irregular than the scaling plots generated by GARCH, the MMAR, and the data. Tables 3 and 4 again complement this visual evidence. The simulated FIGARCH slopes improve over the GARCH slopes, but are not as close to the data as the MMAR. The SSEs from the data are also far in the tails of their distributions generated under FIGARCH.

The previous analysis has separately assessed ten moment conditions that capture different scaling features of the data. We now consider the evidence provided by the global statistics, which are quadratic functions of these ten moment conditions. Each column of the results in table 5 is obtained by a different weighting of the set of quadratic terms, so that within-column comparisons provide four separate views of ability to fit scaling features of the data. Each of the weighting matrices of course has different power against a given model, and asymptotic theory suggests that the most-powerful weighting matrix for each model is provided by the inverse of its own covariance matrix of moment conditions. Thus, we expect the diagonal entries of the table to provide the greatest power to reject each model, and this is consistent with our results. Whether evaluated column-wise or by the diagonal elements, the results confirm that the MMAR is best able to replicate the scaling properties of the data.

These simulations are of course only a preliminary step to evaluating the usefulness of the multifractal model. Nonetheless, our results demonstrate that scaling properties contain important information for estimating and discriminating between models. A natural path for future work will be to incorporate this information in broader estimation and testing procedures. As these techniques develop (for example, Calvet and Fisher (2001)) and lead to more-rigorous eval-

TABLE 3.—MOMENT SCALING IN MODEL SIMULATIONS: ESTIMATED SLOPES

A. Predicted Slopes and Mean Slopes from Model Simulations and DM/USD Data													
q	DM/USD Data	i. MMAR			ii. MMAR ($H = 1/2$)			iii. FIGARCH (1, d , 0)		iv. GARCH (1, 1)		$\bar{\tau}_{sim}$	s.d.
	$\hat{\tau}$	τ_0	$\bar{\tau}_{sim}$	s.d.	τ_0	$\bar{\tau}_{sim}$	s.d.	$\bar{\tau}_{sim}$	s.d.	$\bar{\tau}_{sim}$	s.d.		
0.5	-0.711	-0.71	-0.71	0.01	-0.73	-0.73	0.01	-0.74	0.01	-0.75	0.01		
1.0	-0.440	-0.44	-0.44	0.02	-0.49	-0.48	0.02	-0.49	0.03	-0.50	0.02		
2.0	0.058	0.06	0.05	0.06	0.00	-0.01	0.06	-0.01	0.08	-0.01	0.05		
3.0	0.500	0.49	0.47	0.12	0.43	0.41	0.10	0.42	0.16	0.47	0.08		
5.0	1.208	1.19	1.16	0.27	1.16	1.10	0.24	1.14	0.35	1.36	0.17		
B. Probabilities that the Simulated Model Slopes are Less than the Slopes from DM/USD Data: $\mathbb{P}(\tau_{sim} < \hat{\tau})$													
0.5			0.5910				0.9812			0.9969		0.9993	
1.0			0.5465				0.9596			0.9787		0.9959	
2.0			0.5620				0.8867			0.8363		0.9186	
3.0			0.6032				0.8111			0.7209		0.6587	
5.0			0.5757				0.6771			0.5656		0.1754	
C. Percentiles of the Simulated Model Slopes													
	P0.5	P1	P2.5	P5	P10	P25	P50	P75	P90	P95	P97.5	P99	P99.5
i. MMAR													
0.5	-0.745	-0.742	-0.738	-0.734	-0.729	-0.722	-0.714	-0.705	-0.698	-0.694	-0.690	-0.686	-0.683
1.0	-0.506	-0.501	-0.493	-0.485	-0.475	-0.460	-0.443	-0.426	-0.412	-0.403	-0.396	-0.386	-0.379
2.0	-0.106	-0.091	-0.069	-0.052	-0.028	0.009	0.049	0.090	0.126	0.149	0.170	0.197	0.215
3.0	0.159	0.192	0.246	0.284	0.328	0.399	0.472	0.549	0.622	0.666	0.712	0.758	0.798
5.0	0.343	0.465	0.601	0.705	0.821	0.988	1.162	1.331	1.491	1.595	1.691	1.793	1.871
ii. MMAR ($H = 1/2$)													
0.5	-0.766	-0.762	-0.758	-0.754	-0.750	-0.743	-0.735	-0.727	-0.720	-0.716	-0.712	-0.708	-0.705
1.0	-0.542	-0.536	-0.527	-0.519	-0.511	-0.497	-0.480	-0.465	-0.451	-0.443	-0.436	-0.428	-0.423
2.0	-0.150	-0.136	-0.114	-0.098	-0.078	-0.045	-0.008	0.029	0.062	0.083	0.100	0.124	0.138
3.0	0.143	0.177	0.211	0.244	0.283	0.344	0.410	0.479	0.542	0.584	0.621	0.666	0.707
5.0	0.412	0.502	0.633	0.714	0.807	0.954	1.106	1.258	1.401	1.490	1.581	1.684	1.779
iii. FIGARCH (1, d , 0)													
0.5	-0.776	-0.773	-0.768	-0.764	-0.760	-0.753	-0.744	-0.736	-0.729	-0.725	-0.721	-0.716	-0.713
1.0	-0.566	-0.556	-0.546	-0.537	-0.527	-0.511	-0.494	-0.476	-0.460	-0.450	-0.442	-0.430	-0.422
2.0	-0.244	-0.211	-0.174	-0.141	-0.111	-0.064	-0.016	0.034	0.084	0.115	0.147	0.191	0.221
3.0	-0.082	-0.003	0.086	0.151	0.220	0.320	0.416	0.514	0.608	0.680	0.737	0.815	0.857
5.0	0.033	0.200	0.412	0.559	0.703	0.934	1.155	1.362	1.560	1.699	1.807	1.935	2.031
iv. GARCH (1, 1)													
0.5	-0.777	-0.774	-0.770	-0.766	-0.762	-0.755	-0.747	-0.740	-0.733	-0.729	-0.725	-0.721	-0.718
1.0	-0.555	-0.548	-0.541	-0.534	-0.525	-0.512	-0.497	-0.482	-0.470	-0.461	-0.454	-0.447	-0.442
2.0	-0.121	-0.110	-0.095	-0.080	-0.063	-0.036	-0.005	0.025	0.053	0.071	0.084	0.101	0.113
3.0	0.281	0.302	0.322	0.347	0.375	0.420	0.470	0.521	0.571	0.601	0.629	0.664	0.685
5.0	0.928	0.978	1.037	1.094	1.150	1.250	1.352	1.461	1.569	1.641	1.709	1.790	1.859

This table is based upon $J = 10,000$ simulated paths for each of the four models. In panel A, the column τ_0 is the theoretically predicted slope under the MMAR as calculated in subsection IVB. The column $\bar{\tau}$ shows the average slope over the J paths, and s.d. the standard deviation. Panel B provides the percentage of the simulated paths with slope values less than observed in the daily DM/USD data, and panel C gives percentiles of the simulated distribution under each model.

uations, it will be interesting to discover whether the promise of these early simulation results is fulfilled.

E. Equity Data

After observing multifractal properties in DM/USD exchange rates, it is natural to test the model on other financial data. This section presents evidence of moment scaling in a sample of five major U.S. stocks and one equity index.³⁵

³⁵ The multifractal model offers a flexible framework that may be amenable to many types of financial prices. In particular, equity data

The Center for Research in Security Prices (CRSP) provides daily stock returns for 9,190 trading days from July 1962 to December 1998. We present results for the value-weighted NYSE-AMEX-NASDAQ index ("CRSP Index") and five stocks: Archer Daniels Midland (ADM), General Motors (GM), Lockheed-Martin, Motorola, and United Airlines (UAL). The individual stocks are issued by large,

requires additional consideration for the relationship between volatility and mean returns. Because this has not been incorporated in the version of the model presented in this paper, the results in this section should be interpreted as a model-free investigation of scaling.

TABLE 4.—SUM OF SQUARED ERRORS IN MODEL SIMULATIONS

A. Mean SSE from Simulated Models and DM/USD Data													
q	DM/USD Data	i. MMAR		ii. MMAR ($H = 1/2$)		iii. FIGARCH (1, d , 0)				iv. GARCH (1, 1)			
	SSE	SSE	s.d.	SSE	s.d.	SSE	s.d.	SSE	s.d.				
0.5	0.018	0.0200	0.0080	0.0213	0.00805	0.0241	0.00876			0.0214	0.00779		
1.0	0.0507	0.0604	0.0278	0.0637	0.0266	0.0832	0.0417			0.0586	0.0229		
2.0	0.159	0.330	0.212	0.311	0.167	0.635	0.635			0.21	0.0937		
3.0	0.42	1.49	1.14	1.29	0.864	2.98	2.95			0.698	0.39		
5.0	2.76	10.1	7.13	9.84	6.01	17.3	13.7			4.91	3.54		
B. Probabilities that the Simulated Model SSE are Less than in the DM/USD Data													
0.5		0.4640		0.3914		0.2541				0.3733			
1.0		0.4265		0.3553		0.1648				0.4191			
2.0		0.1084		0.1000		0.0204				0.3211			
3.0		0.0156		0.0197		0.0026				0.1902			
5.0		0.0155		0.0283		0.0034				0.2293			
C. Percentiles of the Simulated Model SSE													
	P0.5	P1	P2.5	P5	P10	P25	P50	P75	P90	P95	P97.5	P99	P99.5
i. MMAR													
0.5	0.00703	0.00774	0.00889	0.0101	0.0116	0.0144	0.0185	0.0238	0.0298	0.0345	0.0394	0.0461	0.0528
1.0	0.0206	0.0223	0.0253	0.0289	0.0331	0.042	0.0546	0.0716	0.0938	0.112	0.131	0.158	0.173
2.0	0.0936	0.103	0.118	0.133	0.156	0.203	0.278	0.392	0.551	0.695	0.87	1.11	1.34
3.0	0.351	0.387	0.457	0.528	0.625	0.834	1.18	1.77	2.65	3.45	4.47	5.86	7.18
5.0	2.23	2.53	2.98	3.44	4.11	5.65	8.18	12.3	18.4	23.4	28.7	36.2	42.5
ii. MMAR ($H = 1/2$)													
0.5	0.00806	0.00882	0.00993	0.0111	0.0127	0.0157	0.0199	0.0253	0.0315	0.0364	0.0412	0.0469	0.0527
1.0	0.0224	0.0246	0.0281	0.0318	0.0362	0.0455	0.0583	0.0759	0.0964	0.112	0.13	0.157	0.175
2.0	0.0973	0.106	0.121	0.138	0.159	0.203	0.271	0.369	0.503	0.617	0.734	0.902	1.08
3.0	0.332	0.372	0.437	0.5	0.583	0.769	1.06	1.52	2.21	2.81	3.55	4.65	5.58
5.0	2.05	2.33	2.67	3.16	3.72	5.02	7.24	10.7	15.7	20.2	24.9	32.1	38.3
iii. FIGARCH (1, d , 0)													
0.5	0.00873	0.00969	0.0112	0.0126	0.0144	0.0179	0.0229	0.0287	0.0355	0.0403	0.0447	0.0507	0.0561
1.0	0.026	0.0291	0.034	0.0386	0.0448	0.0572	0.0745	0.0984	0.129	0.155	0.185	0.228	0.276
2.0	0.126	0.14	0.165	0.192	0.227	0.31	0.461	0.73	1.18	1.62	2.14	3.1	4.18
3.0	0.453	0.527	0.64	0.751	0.931	1.34	2.11	3.6	5.75	7.74	10.3	14.3	19.4
5.0	2.99	3.42	4.13	4.94	6.07	8.73	13.5	21.4	32.1	41.5	52.4	68.9	85.3
iv. GARCH (1, 1)													
0.5	0.00804	0.00872	0.00995	0.0111	0.0128	0.0159	0.0202	0.0254	0.0315	0.0357	0.0397	0.0455	0.05
1.0	0.0212	0.0232	0.0265	0.0299	0.0343	0.0427	0.0546	0.0697	0.0874	0.101	0.115	0.135	0.149
2.0	0.0725	0.0793	0.091	0.103	0.116	0.147	0.19	0.249	0.325	0.383	0.449	0.548	0.624
3.0	0.214	0.237	0.27	0.31	0.356	0.456	0.608	0.827	1.13	1.39	1.64	2.07	2.46
5.0	1.25	1.39	1.59	1.83	2.14	2.84	3.95	5.83	8.5	11.1	13.9	18.4	23.2

This table is based upon the same simulated paths as table 3. In panel A, the column SSE shows the average SSE over the $J = 10,000$ paths for each model, and s.d. gives the standard deviation. Panel B gives the percentage of simulated paths with SSE less than in the daily DM/USD data, and panel C provides percentiles of the simulated SSE under each model.

well-known corporations from various economic sectors, and have reported data for the full CRSP sample span.³⁶ For each series, we convert the daily return data into a renormalized log-price series X_t , and then apply the partition function methodology described in subsection VIB.³⁷

Figure 10 shows results for the CRSP index and GM. In the first two panels, the full data sets are used with

increments Δt ranging from one day to approximately one year. The partition functions for moments $q = \{1, 2, 3\}$ are approximately linear for both series, with little variation around the apparent slope. The slope for the moment $q = 2$ is noticeably positive for the CRSP index, indicating persistence. This characteristic is very atypical of individual securities, although short-horizon persistence has been observed previously in index returns, and is typically attributed to asynchronous trading (Boudoukh, Richardson, & Whitelaw, 1994). In contrast to the results for low moments, the partition functions from both series vary considerably for the moment $q = 5$. This

³⁶ Choosing stocks with full samples allows testing of the moment-scaling restrictions over a larger range of frequencies.

³⁷ The CRSP holding period returns $r_t = (P_t - P_{t-1} + d_t)/P_{t-1}$ include cash distributions d_t . We construct the series $\{X_t\}_{t=0}^T$ by $X_0 = 0$, $X_t = X_{t-1} + \ln(1 + r_t)$.

TABLE 5.—GLOBAL TESTS FOR REPLICATION OF DM/USD SCALING

Model	Weighting			
	W_1	W_2	W_3	W_4
	Test Statistics and Simulated P-values			
MMAR	7.57 (0.5219)	10.23 (0.4452)	7.26 (0.5539)	27.55 (0.3215)
MMAR ($H = 1/2$)	10.79 (0.2133)	12.46 (0.2117)	11.75 (0.2099)	19.69 (0.2905)
FIGARCH (1, d , 0)	24.60 (0.1159)	29.29 (0.1605)	25.96 (0.0427)	58.79 (0.2586)
GARCH (1, 1)	21.05 (0.0274)	22.95 (0.0239)	25.34 (0.0297)	27.04 (0.0324)

The weighting matrices $W_1, \dots, W_4$ correspond to the inverse covariance matrix of the moment conditions under the four models: MMAR, MMAR ($H = 1/2$), FIGARCH, and GARCH. The ten moment conditions used are the expected slope and SSE statistics for the five moments $q = 0.5, 1, 2, 3, 5$, conditioned upon each model. The expectations and covariances of the moment conditions are obtained by simulating 10,000 paths under each model. The statistics reported are from the daily DM/USD data, and the P -values show the probability of observing a larger statistic conditioning upon the assumed model. Asymptotic theory suggests that the diagonal entries will have most power against each model.

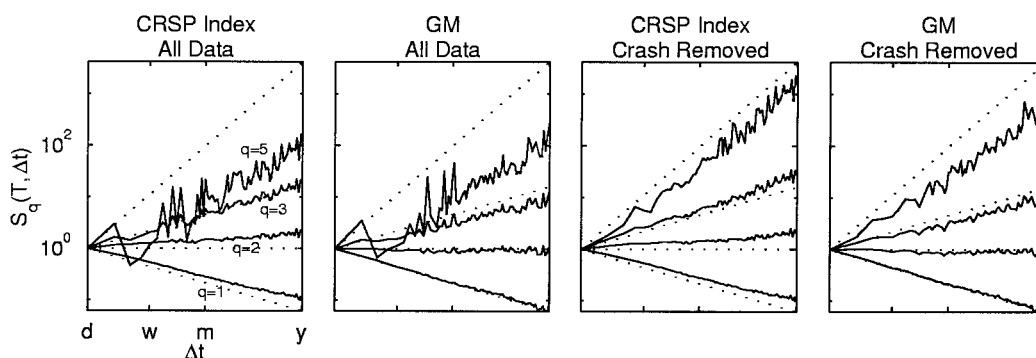
suggests investigation of the tails of the data. We find that the behavior of the fifth moment is dominated by volatility surrounding the stock market crash of October 1987. This is demonstrated by the second two panels of figure 10, which show striking linearity after simply removing the crash day from both data sets.

Because discarding outliers seems an unsatisfactory approach to volatility modeling,³⁸ we reexamine the full data sets. The partition functions $S_{q=5}(T, \Delta t)$ for both series drop considerably from $\Delta t = 2$ days to $\Delta t = 3$ days. In the raw data, the CRSP index falls 17% on the day of the crash, but rebounds more than 8% two days later. GM loses 21% in the crash, recovering almost all its losses over the next two days. When $\Delta t = 3$ days, aggregation of these returns within a single interval contributes to the severe declines in the $q = 5$ partition functions.³⁹ As Δt grows, the crash and the two following days occasionally fall in separate intervals and the partition functions spike. More often, however, these three days lie within a single interval when Δt is large. Moreover, when a spike does occur for this reason with large Δt , its size is smaller because of the diminishing

³⁸ Although discarding data may be justified in specific circumstances, our approach in this paper has been to build a stationary model that is flexible enough to accommodate a wide range of changing economic circumstances. This includes both long-range structural shifts and extreme tail events such as the 1987 crash.

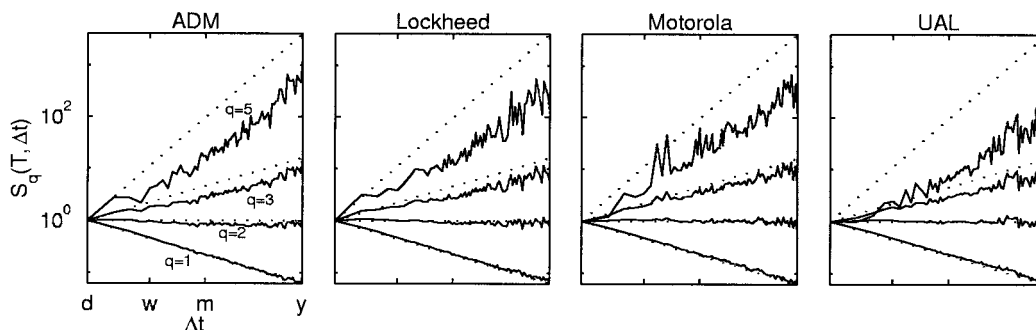
³⁹ When calculating the partition functions for $\Delta t = 2$ days, the crash day and the two following days belong to separate intervals, each of which make large contributions to $S_5(T, \Delta t = 2$ days). When $\Delta t = 3$ days, however, the crash and the two following days belong to a single interval. Their returns partially cancel when aggregated, explaining the decreases in the partition functions $S_5(T, \Delta t)$ at $\Delta t = 3$ days.

FIGURE 10.—PARTITION FUNCTIONS FOR CRSP INDEX AND GM



The data spans from 1962 to 1998, and the time increments labeled “d,” “w,” “m,” and “y” correspond to one day, one week, one month, and one year, respectively. When the full data sets are used, we observe scaling in the first three moments, and for horizons $\Delta t \geq 3$ days in the fifth moments. The drop in the fifth moments between two and three days is caused by sharp rebounds for both series from the 1987 crash. After removing the crash from the data, the second two panels show striking linearity, but change the slopes of the full-sample plots to increase and thus appear more Brownian.

FIGURE 11.—PARTITION FUNCTIONS FOR ADM, LOCKHEED, MOTOROLA, AND UAL



Each of these show strong scaling properties despite the 1987 crash.

influence of the crash at low frequencies. This discussion explains why the partition functions $S_5(T, \Delta t)$ show large variability, but primarily for low values of Δt . Because of this variability, the estimated slopes of the partition function would appear to be relatively imprecise.

After removing the crash from both data sets, the same partition functions appear to give a more-precise fit, but at a cost. Both slopes increase to appear more Brownian or “mild,” suggesting that important information has been lost. Additionally, removing the crash does not necessarily improve the fit in the MMAR because the theoretically predicted slopes constrain only the expectations of the partition functions, not their variability.⁴⁰ In fact, the simulations in the previous section indicate that multifractal paths often have partition functions that vary considerable around their expected slope. Thus, removing the crash gives a false impression of improved model fit and alters scaling properties to imply a much milder process.

The other four stocks in our sample scale remarkably well despite the crash, as shown in figure 11. Consistent with the martingale hypothesis for returns, three of the four stocks have almost exactly flat partition functions for $q = 2$, whereas ADM has a slight negative slope. The difference between Brownian scaling and multiscaling becomes perceptible for $q = 3$, and, for the fifth moment, this difference is pronounced. UAL appears the most variable, with lower slopes at higher moments and thus a wider multifractal spectrum.

Although not exhaustive, our empirical analysis indicates that moment scaling is a prominent feature of many financial series. Using DM/USD data, we confirm this property across three orders of magnitude of frequencies and 23 years of daily returns. A simple estimation procedure helps to provide a specification of the multifractal model that reproduces scaling patterns found in the data. Finally, our analysis of equity data shows that the partition function plots summarize a great deal of information in a convenient form. This new tool may thus be useful in uncovering empirical regularities and building new financial models.

VII. Conclusion

The multifractal model of asset returns (MMAR) is a continuous-time stochastic process that incorporates the outliers and volatility persistence exhibited by many financial time series. The model compounds a standard Brownian motion with an independent, multifractal, time-deformation process that produces volatility clustering. We show how to construct a class of candidate time deformations as the limit of a simple iterative procedure, called a *multiplicative cascade*. The cascade provides parsimonious modeling and results in a generalized scaling rule that restricts return

moments to vary as power laws of the time increment. The price process is a semi-martingale with uncorrelated returns, and thus precludes arbitrage in a standard two-asset setting.

The MMAR offers a fundamentally new class of processes to both finance and mathematics. Multifractal processes have continuous sample paths, but lie outside the class of Itô diffusions. Whereas standard processes can be characterized by a single local scale that describes the local growth rate of variation, sample paths of multifractal processes contain a continuum of local Hölder exponents within any time interval. The distribution of these exponents is conveniently quantified by a renormalized density, the multifractal spectrum $f(\alpha)$. For a large class of multifractal processes, the spectrum can be explicitly derived from Cramér’s large-deviation theory. We demonstrate through a number of examples the sensitivity of the multifractal spectrum to the generating mechanism. The applied researcher may thus relate an empirical estimate of the spectrum back to a particular construction of the process, and is permitted considerable flexibility in modeling different types of data.

We find evidence of multifractality in the moment-scaling behavior of Deutsche mark/U.S. dollar exchange rates. Over a range of observational frequencies from approximately two hours to 180 days, and over a range of time from 1973 to 1996, moments of the data grow approximately like a power law. We obtain an estimate of the multifractal spectrum by a Legendre transform of the moments’ growth rates. From the shape of the estimated spectrum, we infer a log normal distribution as the primitive of the generating mechanism, and estimate its parameters. We simulate the process and confirm that the multifractal model replicates the moment behavior found in the data. We also demonstrate scaling behavior in an equity index and five major U.S. stocks.

Our results indicate several directions for future research. Using our simulation results as a guide, the moment-scaling features of the data can be incorporated into broader estimation and testing procedures. Risk analysis, forecasting, and option pricing are promising applications that are currently being developed in other papers. Further research will also seek to derive the MMAR as an equilibrium process of economies with fully rational agents. In such frameworks, multifractality is expected to arise in equilibrium either exogenously, for instance as a consequence of multifractal technological shocks, or endogenously because of market incompleteness or informational cascades. The early empirical success of the MMAR thus offers new challenges in econometrics, finance, and economic theory.

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⁴⁰ The simulations in the previous section do, however, suggest that the variability of partition function plots can usefully be incorporated into estimation.

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APPENDIX

A. Scaling Rule

This appendix analyzes the set $\mathcal{D}$ defined by a multiplicative measure with parameter $b \geq 2$. Consider a fixed instant $t \in [0, 1]$. For all $\epsilon > 0$, there exists a dyadic number t_n such that $|t_n - t| < \epsilon$. We can then find a number $\Delta_n = b^{-k_n} < \epsilon$ for which $(t_n, \Delta_n) \in \mathcal{D}$. In the plane $\mathbb{R}^2$, the point $(t, 0)$ is thus the limit of the sequence $(t_n, \Delta_n) \in \mathcal{D}$, which establishes

Property 1: The closure of $\mathcal{D}$ contains the set $[0, 1] \times \{0\}$. The scaling relation (4) thus holds "in the neighborhood of any instant."

B. Proof of Proposition 1

Consider two exponents, q_1 and q_2 , and two positive weights, w_1 and w_2 , adding up to 1. Hölder's inequality implies

$$\mathbb{E}(|X(t)|^q) \leq [\mathbb{E}(|X(t)|^{q_1})]^{w_1} [\mathbb{E}(|X(t)|^{q_2})]^{w_2},$$

where $q = w_1 q_1 + w_2 q_2$. Taking logarithms and using equation (5), we obtain

$$\begin{aligned} \ln c(q) + \tau(q) \ln t &\leq [w_1 \tau(q_1) + w_2 \tau(q_2)] \ln t + [w_1 \ln c(q_1) \\ &\quad + w_2 \ln c(q_2)]. \end{aligned} \quad (\text{A1})$$

We divide by $\ln t < 0$, and let t go to zero:

$$\tau(q) \geq w_1 \tau(q_1) + w_2 \tau(q_2), \quad (\text{A2})$$

which establishes the concavity of τ . This proof also contains additional information on multifractal processes. Assuming that relation (5) holds for $t \in [0, \infty)$, we divide inequality (A1) by $\ln t > 0$ and let t go to infinity. We obtain the reverse of inequality (A2), and conclude that $\tau(q)$ is linear. Thus, exact multiscaling can hold only for bounded time intervals $\mathcal{T}$.

C. Proof of Theorem 1

Because the trading time and the Brownian motion $B(t)$ are independent, conditioning on $\theta(t)$ yields

$$\begin{aligned} \mathbb{E}\{|X(t)|^q | \theta(t) = u\} &= \mathbb{E}[|B(u)|^q | \theta(t) = u] \\ &= \theta(t)^{q/2} \mathbb{E}[|B(1)|^q], \end{aligned}$$

and thus $\mathbb{E}[|X(t)|^q] = \mathbb{E}[\theta(t)^{q/2}] \mathbb{E}[|B(1)|^q]$. The process $X(t)$ satisfies the multiscaling relation (5), with $\tau_X(q) \equiv \tau_\theta(q/2)$ and $c_X(q) \equiv c_\theta(q/2) \mathbb{E}[|B(1)|^q]$.

D. Proof of Theorem 2

Let $\mathcal{F}_t$ and $\mathcal{F}'_t$ denote the natural filtrations of $\{X(t)\}$ and $\{X(t), \theta(t)\}$. For any t, T, u , the independence of B and θ implies

$$\begin{aligned} \mathbb{E}\{X(t+T) | \mathcal{F}'_t, \theta(t+T) = u\} &= \mathbb{E}\{B(u) | \mathcal{F}'_t\} \\ &= B[\theta(t)] \end{aligned}$$

because $\{B(t)\}$ is a martingale. We now infer that $\mathbb{E}[X(t+T) | \mathcal{F}_t] = X(t)$, which establishes that $X(t)$ is a martingale and has thus uncorrelated increments. The price $P(t)$ is a smooth function of $X(t)$ and therefore a semi-martingale, which precludes arbitrage opportunities in the two-asset economy.

E. Proof of Theorem 4

Trading time

Consider a canonical cascade after $k \geq 1$ stages. Consistent with the notation of section II, the interval $[0, T]$ is partitioned into cells of length $\Delta t = b^{-k}T$, and $I_1 = [t_1, t_1 + \Delta t]$ and $I_2 = [t_2, t_2 + \Delta t]$ denote two distinct cells with lower endpoints of the form $t_l/T = 0.\eta_1 \dots \eta_k$ and $t_2/T = 0.\zeta_1 \dots \zeta_k$. Assume that the first $l \geq 1$ terms are equal in the b -adic expansions of t_1/T and t_2/T , so that $\zeta_1 = \eta_1, \dots, \zeta_l = \eta_l$, and $\zeta_{l+1} \neq \eta_{l+1}$. The distance $t = |t_2 - t_1|$ satisfies $b^{-l-1} \leq t/T < b^{-l}$, and the product $\mu(I_1)^q \mu(I_2)^q$, which is equal to

$$\Omega_{\eta_1, \dots, \eta_k}^q \Omega_{\zeta_1, \dots, \zeta_k}^q (M_{\eta_1}^{2q} \dots M_{\eta_l}^{2q}) (M_{\eta_1, \dots, \eta_{l+1}}^q \dots M_{\eta_1, \dots, \eta_k}^q) (M_{\zeta_1, \dots, \zeta_{l+1}}^q \dots M_{\zeta_1, \dots, \zeta_k}^q),$$

has mean $(\mathbb{E} \Omega^q)^2 [\mathbb{E} M^{2q}]^l [\mathbb{E} M^q]^{2(k-l)}$. We conclude that

$$\begin{aligned} \text{Cov} [\mu(I_1)^q; \mu(I_2)^q] &= (\mathbb{E} \Omega^q)^2 (\mathbb{E} M^q)^{2k} \{[(\mathbb{E} M^{2q})/(\mathbb{E} M^q)^2]^l - 1\} \\ &= C_1(\Delta t)^{2\tau_0(q)+2} [b^{-l(\tau_0(2q)-2\tau_0(q)-1)} - 1] \end{aligned}$$

is bounded by two hyperbolic functions of t .

Log-Price

Because $B(t)$ and $\theta(t)$ are independent processes, the conditional expectation

$$\mathbb{E}[|X(0, \Delta t) X(t, \Delta t)|^q | \theta(\Delta t) = u_1, \theta(t) = u_2, \theta(t + \Delta t) = u_3], \quad (\text{A3})$$

simplifies to

$$\mathbb{E}[|B(u_1)|^q] \mathbb{E}[|B(u_3) - B(u_2)|^q] = |u_1|^{q/2} |u_3 - u_2|^{q/2} [\mathbb{E}|B(1)|^q]^2.$$

Taking expectations, we infer that

$$\mathbb{E}[|X(0, \Delta t) X(t, \Delta t)|^q] = \mathbb{E}[|\theta(0, \Delta t) \theta(t, \Delta t)|^{q/2}] [\mathbb{E}|B(1)|^q]^2$$

and therefore

$$\delta_X(t, q) = \delta_\theta(t, q/2) [\mathbb{E}|B(1)|^q]^2.$$

F. Interpretation of $f(\alpha)$ as a Fractal Dimension

Fractal geometry considers irregular and winding structures that are not well described by their Euclidean length. For instance, a geographer measuring the length of a coastline will find very different results as she increases the precision of her measurement. In fact, the structure of the coastline is usually so intricate that the measured length diverges to infinity as the geographer's measurement scale goes to zero. For this reason, we cannot use the Euclidean length to compare two different coastlines, and it is natural to introduce a new concept of dimension. Given a precision level $\epsilon > 0$, we consider coverings of the coastline with balls of diameter ϵ . Let $N(\epsilon)$ denote the smallest number of balls required for such a covering. The approximate length of the coastline is defined by $L(\epsilon) = \epsilon N(\epsilon)$. In many cases, $N(\epsilon)$ satisfies a power law as ϵ goes to zero, $N(\epsilon) \sim \epsilon^{-D}$, where D is a constant called the fractal dimension. Fractal dimension helps to analyze the structure of a fixed multifractal. For any $\alpha \geq 0$, we can define the set $T(\alpha)$ of instants with Hölder exponent α . As any subset of the real line, $T(\alpha)$ has a fractal dimension $D(\alpha)$, which satisfies $0 \leq D(\alpha) \leq 1$. It can be shown that, for a large class of multifractals, the dimension $D(\alpha)$ coincides with the multifractal spectrum $f(\alpha)$.

In the case of measures, we can provide a heuristic interpretation of this result based on coarse Hölder exponents. Denoting by $N(\alpha, \Delta t)$ the number of intervals $[t, t + \Delta t]$ required to cover $T(\alpha)$, we infer from equation (1) that $N(\alpha, \Delta t) \sim (\Delta t)^{-f(\alpha)}$. We then rewrite the total mass

$\mu[0, T] = \sum \mu(\Delta t) \sim \sum (\Delta t)^{\alpha(t)}$, and rearrange it as a sum over Hölder exponents:

$$\mu[0, T] \sim \int (\Delta t)^{\alpha-f(\alpha)} d\alpha.$$

The integral is dominated by the contribution of the Hölder exponent α_1 that minimizes $\alpha - f(\alpha)$, and therefore

$$\mu[0, T] \sim (\Delta t)^{\alpha_1 - f(\alpha_1)}.$$

Because the total mass $\mu[0, T]$ is positive, we infer that $f(\alpha_1) = \alpha_1$, and $f(\alpha) \leq \alpha$ for all α . When f is differentiable, the coefficient α_1 also satisfies $f'(\alpha_1) = 1$. The spectrum $f(\alpha)$ then lies under the 45-degree line, with tangential contact at $\alpha = \alpha_1$.

G. Large-Deviation Theory and the Multifractal Spectrum

This section sketches the proof of Theorem 6 and introduces the concepts of latent and virtual Hölder exponents.⁴¹ First consider a conservative multiplicative measure μ . Application of large-deviation theory (LDT) begins with the histogram method of subsection IVA. Subdivide the range of α 's into intervals of length $\Delta\alpha$, and denote by $N_k(\alpha)$ the number of coarse Hölder exponents in the interval $(\alpha, \alpha + \Delta\alpha]$. For large values of k , we write

$$\frac{1}{k} \log_b \left[\frac{N_k(\alpha)}{b^k} \right] \sim \frac{1}{k} \log_b \mathbb{P}\{\alpha < \alpha_k \leq \alpha + \Delta\alpha\}. \quad (\text{A4})$$

This relation holds exactly for multinomial measures, which have discrete coarse exponents α_k , but is postulated in more-general cases. For any $\alpha > \alpha_0$, Cramér's theorem implies that

$$k^{-1} \log_b \mathbb{P}\{\alpha_k > \alpha\} \rightarrow \inf_q \log_b [\mathbb{E} e^{q(\alpha - V_1)} \ln b] \quad (\text{A5})$$

as $k \rightarrow \infty$. Using the definition of the scaling function, we simplify the limit to $\inf_q [\alpha q - \tau(q)] - 1$. Combining this with equations (1) and (A4), it follows⁴² that Theorem 6 holds.

These arguments easily extend to a canonical measure μ . Given a b -adic instant t , the coarse exponent $\alpha_k(t) = \ln \mu[t, t + \Delta t] / \ln \Delta t$ is the sum of a high-frequency component, $-k^{-1} \log_b \Omega_{\eta_1, \dots, \eta_k}$, and of the familiar low-frequency average

$$\alpha_{k,L}(t) = -[\log_b M_{\eta_1} + \dots + \log_b M_{\eta_1, \dots, \eta_k}] / k.$$

The exponent $\alpha_k(t)$ converges almost surely to $\alpha_0 = -\mathbb{E} \log_b M$, and the multifractal spectrum is again the Legendre transform of the scaling function $\tau(q)$.

Relation (A5) also shows that $f(\alpha)$ is the limit of

$$k^{-1} \log_b \mathbb{P}\{\alpha_{k,L}(t) > \alpha\} + 1 \quad \text{if } \alpha > \alpha_0,$$

and

$$k^{-1} \log_b \mathbb{P}\{\alpha_{k,L}(t) < \alpha\} + 1 \quad \text{if } \alpha < \alpha_0.$$

$f(\alpha)$ is therefore a hump-shaped function, reaching a maximum at the most-probable exponent: $f(\alpha) \leq f(\alpha_0) = 1$.⁴³ We have successively viewed the spectrum $f(\alpha)$ as:

⁴¹ We refer the reader to Mandelbrot (1989b), Peyrière (1992), and CFM (1997) for more-detailed discussions.

⁴² See CFM (1997) for a more-detailed proof.

⁴³ It is easy to show that $\alpha_0 q - \tau(q)$ is minimal for $q = 0$. The set $T(\alpha_0)$ has therefore fractal dimension $f(\alpha_0) = -\tau(0) = 1$, and thus carries all of the Lebesgue measure. Moreover, by the central limit theorem, $f(\alpha)$ is locally quadratic around α_0 , as shown by CFM (1997).

- (D1) the limit of a renormalized histogram of coarse Hölder exponents,
- (D2) the fractal dimension of the set of instants with Hölder exponent α , and
- (D3) the limit of $k^{-1} \log_b \mathbb{P}\{\alpha_{k,L}(t) > \alpha\} + 1$ provided by LDT.

The three definitions coincide for multinomial measures, and (D1) and (D2) agree for a large class of multifractals (Peyrière, 1991). However, (D1) and (D2) imply that $f(\alpha) \geq 0$, whereas (D3) imposes no such restriction. When $f(\alpha) < 0$, the corresponding α 's, called latent, are rare coarse exponents, which appear in few draws of the random measure and control high and low moments (Mandelbrot, 1989b). Similarly, because canonical measures allow M to be greater than 1, the low-frequency average $\alpha_{k,L}(t)$ can be negative with positive probability. (D3) thus defines the multifractal spectrum for negative, or virtual, values of α . This topic, further discussed by Mandelbrot (1989b), remains an active research area in mathematics.

H. Proof of Theorem 7

Given a process Z , denote $\alpha_Z(t)$ as its local scale at date t , and $T_Z(\alpha)$ as the set of instants with scale α . At any date, the infinitesimal variation of the log-price $X(t + \Delta t) - X(t) = B[\theta(t + \Delta t)] - B[\theta(t)]$ satisfies

$$|X(t + \Delta t) - X(t)| \sim |\theta(t + \Delta t) - \theta(t)|^{1/2} \sim |\Delta t|^{\alpha_\theta(t)/2},$$

implying $\alpha_X(t) \equiv \alpha_\theta(t)/2$. The sets $T_X(\alpha)$ and $T_\theta(2\alpha)$ coincide, and in particular have identical fractal dimensions: $f_X(\alpha) \equiv f_\theta(2\alpha)$. Moreover, because the price $P(t)$ is a differentiable function of $X(t)$, the two processes have identical local Hölder exponents and spectra.

I. Simulation of Multifractal Paths

The construction of the simulated multifractal price paths used in subsection VID has two basic components. First, a finite-stage approximation to a canonical multifractal measure is constructed as suggested in subsection IIB. All simulations use the base $b = 2$, so that, if a simulation of length T is desired, we choose the minimum integer number of stages k such that $2^k \geq T$. At each stage in the construction, we draw independent log normal multipliers with identical distributions given by the results in subsection VIC. When k stages are completed, the measure μ_k is used as a discrete approximation to the quadratic variation of a multifractal path. Aggregating the increments of μ_k thus provides a simulated path from the trading time $\theta(t)$. The second part of the construction involves compounding, as suggested by Assumptions 1 through 3 of section III. For the martingale version of the MMAR, we simply calculate the standard deviation $[\mu_k(\Delta t)]^{1/2}$ over the discrete-time interval Δt and multiply by an independent standard Gaussian. To simulate the extended MMAR, we first generate a discretized path from an FBM with parameter $\hat{H}$ taken from the estimates in subsection VIC. Interpolation provides values of the path $B_H[\theta(t)]$ at the simulated values from the path $\theta(t)$.